RAPIDe-3

Study Title:	A Phase 3, Randomized, Double-blind, Placebo-controlled, Cross-over Study of Oral Deucrictibant Soft Capsule for On-Demand Treatment of Attacks in Adolescents and Adults with Hereditary Angioedema
Acronym:	RAPIDe-3
Study Number:	PHA022121-C306
Study Phase:	Phase 3
Product Name:	Deucrictibant soft capsule
Indication:	Hereditary Angioedema
EU Trial Number:	2023-507268-37
IND Number:	155872
Investigators:	Multicenter
Sponsor:	Pharvaris Netherlands B.V. J.H. Oortweg 21 2333 CH Leiden The Netherlands
Sponsor Contact:	clinical@pharvaris.com
Sponsor Representative(s):	[redacted] Chief Medical Officer Pharvaris Netherlands B.V. [redacted] Clinical Development Lead Vice President Pharvaris Netherlands B.V.

Protocol Version	**Date**
Original Protocol:	03 November 2023
Amendment 1.0:	01 February 2024
Amendment 1.1 (EU only):	21 June 2024
Amendment 1.2 (EU only):	03 July 2024
Amendment 1.3 (ZA only):	26 July 2024
Amendment 2.0:	19 May 2024
Amendment 3.0:	28 October 2024
Amendment 4.0:	15 January 2025
Amendment 4.1 (EU only):	17 January 2025

This clinical study is to be conducted in compliance with the protocol, with the Regulation (EU) No 536/2014, and with the principles of Good Clinical Practice.

SPONSOR SIGNATURE PAGE

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Study Number:	PHA022121-C306
Amendment 4.1:	17 January 2025

This clinical study protocol was subject to critical review and has been approved by the Sponsor.

Signed: [redacted] Date: _____

[redacted]
Clinical Development Lead, Vice President
Pharvaris Netherlands B.V.

INVESTIGATOR'S AGREEMENT

Study Title:	A Phase 3, Randomized, Double-blind, Placebo-controlled, Cross-over Study of Oral Deucrictibant Soft Capsule for On-Demand Treatment of Attacks in Adolescents and Adults with Hereditary Angioedema
Study Number:	PHA022121-C306
Amendment 4.1:	17 January 2025

I have read all pages of this clinical study protocol for which Pharvaris Netherlands B.V. is the sponsor. I agree that it contains all the information required to conduct this study. I agree to conduct the study as outlined in the protocol and to comply with all the terms and conditions set out therein. I confirm that I will conduct the study in accordance with the International Council for Harmonisation (ICH) GCP guidelines (ICH E6[R2]) and the provisions of the Helsinki Declaration. I will also ensure that all relevant members of my staff have access to copies of this protocol, the ICH GCP guidelines, and the Helsinki Declaration to enable them to work in accordance with the provisions of these documents."

Signed:_____ Date:_____

Name:_____

<enter name and credentials>
<enter title>
<enter affiliation>
<enter address>
<enter phone number>

STUDY ADMINISTRATIVE STRUCTURE

Table 1: Study Administrative Structure

Investigators:	Names and contact details of participating investigators are available in the electronic Trial Master File (eTMF)	
Sponsor:	Pharvaris Netherlands B.V. J.H. Oortweg 21 2333 CH Leiden The Netherlands	
Sponsor Representatives:	[redacted] Chief Medical Officer	[redacted] Clinical Development Lead, Vice President
Medical Monitor:	Medpace Inc. (Cincinnati, Ohio, USA)	

1. SYNOPSIS

Name of Sponsor/Company: Pharvaris Netherlands B.V. (Leiden, The Netherlands)		
Name of Investigational Product: Deucrictibant soft capsule		
Name of Active Ingredient: Deucrictibant		
Protocol Number: PHA022121-C306	**Phase:** 3	**Country:** International
Title of Study: A Phase 3, Randomized, Double-blind, Placebo-controlled, Cross-over Study of Oral Deucrictibant Soft Capsule for On-Demand Treatment of Attacks in Adolescents and Adults with Hereditary Angioedema		
EU Trial Number: 2023-507268-37		

Objectives:

Primary:

- To evaluate the efficacy of deucrictibant as on-demand treatment compared with placebo on the onset of symptom relief during hereditary angioedema (HAE) attacks

Secondary:

- To evaluate the efficacy of deucrictibant as on-demand treatment compared with placebo on symptom relief and resolution of HAE attacks
- To evaluate the safety and tolerability of deucrictibant compared with placebo for on-demand treatment of HAE attacks
- To assess the pharmacokinetics (PK) of deucrictibant in adolescent participants (≥12 to <18 years) in a non-attack state

Exploratory:

- To evaluate participants' quality of life (QoL)

Study Endpoints:

Primary Efficacy Endpoint

- Time to onset of symptom relief, defined as Patient Global Impression of Change (PGI-C) rating of at least "a little better" for 2 consecutive timepoints within 12 hours post-treatment

Secondary Efficacy Endpoints

1. Proportion of study drug-treated attacks achieving PGI-C rating of at least "a little better" at 4 hours post-treatment

2. Time to substantial symptom relief, defined as achieving PGI-C rating of at least "better" for 2 consecutive timepoints within 12 hours post-treatment
3. Time to substantial symptom relief by Patient Global Impression of Severity (PGI-S), defined as achieving ≥1 point reduction in PGI-S from pre-treatment for 2 consecutive timepoints within 12 hours post-treatment
4. Time to complete symptom resolution, defined as achieving PGI-S rating of "none" within 48 hours post-treatment
5. Time to End of Progression (EoP) in attack symptoms within 12 hours, with EoP time defined as the earliest post-treatment timepoint after which all subsequent PGI-C ratings are stable or improved
6. Proportion of study drug-treated attacks requiring rescue medication within 24 hours post-treatment
7. Proportion of attacks achieving symptom resolution, defined as achieving PGI-S rating of "none" with one dose of study drug at 24 hours post-treatment
8. Time to substantial symptom relief by Angioedema SyMptom Rating scAle (AMRA), defined as a ≥50% reduction in AMRA composite score from pre-treatment for 2 consecutive timepoints within 12 hours post-treatment
9. Time to almost complete or complete symptom relief by AMRA, defined as all item scores in AMRA having a value ≤10 for 2 consecutive timepoints within 24 hours post-treatment
10. Proportion of study drug-treated attacks reaching almost complete or complete symptom relief by AMRA, defined as all item scores in AMRA having a value ≤10 at 24 hours post-treatment
11. Time to EoP in attack symptoms within 12 hours, with EoP time defined as the earliest post-treatment timepoint after which every individual AMRA item is stable or improved at all subsequent timepoints

Safety Endpoints

- Incidence of treatment-emergent adverse events (TEAEs) and serious AEs (SAEs)
- Change from baseline in clinical laboratory, vital sign, and electrocardiogram (ECG) parameters

Pharmacokinetic Endpoints (adolescents only)

- Deucrictibant plasma concentration-time profiles
- PK parameters

Exploratory Endpoint

- Changes in health-related quality of life (QoL)

Study Design:

This is a Phase 3, multicenter, randomized, double-blind, placebo-controlled, 2-period, 2-treatment cross-over study to evaluate the efficacy and safety of orally administered deucrictibant for the on-

demand treatment of attacks in participants ≥12 to ≤75 years of age with HAE (type 1, type 2, or type 3), a proportion of whom are using long-term prophylactic medication for HAE.

The study consists of a Screening Phase during which eligibility is confirmed, a Treatment Phase in which participants will be randomized and receive double-blinded study drug to treat 2 qualified HAE attacks (i.e., 2 Treatment Periods within the Treatment Phase) at home, and an End-of-Study Follow-up Phase after the second study drug-treated attack. In addition, for adolescent participants (age ≥12 to <18 years), PK samples are collected after administration of deucrictibant at Day 1 in a non-attack state.

After signing the informed consent form (ICF)/assent form, participants will be screened for eligibility. During the Screening Phase, participants will attend an on-site visit to confirm eligibility, to undergo screening procedures, and to receive training on the use of the electronic device with the electronic patient-reported outcome (ePRO) instruments and the electronic HAE diary. The participants will complete an ePRO competency assessment (attack simulation) to ensure their ability to complete required ePRO assessments and questionnaires with the electronic device during an actual attack during the treatment phase.

After confirmation of eligibility, participants will be randomized on Day 1 to a crossover treatment sequence of double-blinded study drug in a 1:1 ratio (deucrictibant 20 mg for the first attack, placebo for the second attack or the reverse order) to treat 2 qualifying attacks, where each attack could be either a qualifying non-laryngeal attack or, in adult participants only, a non-severe laryngeal attack not associated with breathing difficulties or stridor, during the Treatment Phase.

Randomization will be stratified according to participant age (≥12 to <18 years, ≥18 years) and use of long-term prophylaxis HAE therapy (Yes/No). The study population will include approximately 30% of participants who are using long-term prophylactic medication for HAE globally; in the United States (including Puerto Rico), enrollment is targeted to be approximately 50% of participants using long-term prophylactic medication.

Refer to Section 6.1.2.3.1 for details regarding attack qualification by the investigator (or designee) and treatment, based on attack type.

After the participant self-administers study drug for the first qualifying HAE attack, they will have either an on-site or remote Post-Attack Visit ≥48 hours to ≤10 days after study drug administration for evaluation and recording of adverse events (AEs) and concomitant medication use.

For study drug-treated **non-laryngeal attacks**, a second dose of study drug is permitted ≥4 hours post first dose in case symptoms are persisting or progressing after the first study drug dose. If at ≥1 hour after the second study drug dose, symptoms still persist or progress, then HAE rescue medication may be administered. Note that, if needed for participant safety, participants may self-administer HAE rescue medication at any time. The time of administration of study drug as well as HAE rescue medication must be recorded in the electronic device.

For study drug-treated **non-severe laryngeal attacks (in adults only)**, the participant may self-administer rescue medication at any time, if needed. A second dose of study drug should not be administered for non-severe laryngeal attacks.

Laryngeal attacks, regardless of severity, in adolescent participants or severe laryngeal attacks, including those associated with breathing difficulties or stridor, in adult participants should not be treated with study drug, and should be treated immediately with HAE stand-of-care medication and medical care must be sought urgently.

After the participant self-administers study drug for the second qualifying HAE attack, an on-site End-of-Study Visit is to occur at 10 (±5) days following attack resolution.

For participants who discontinue from the study prematurely, the End-of-Study Visit will be performed within 10 (±5) days after the decision to discontinue the study is communicated to the Investigator. Participants who withdraw consent for further participation will be asked to attend the End-of-Study Visit as soon as possible for safety assessments.

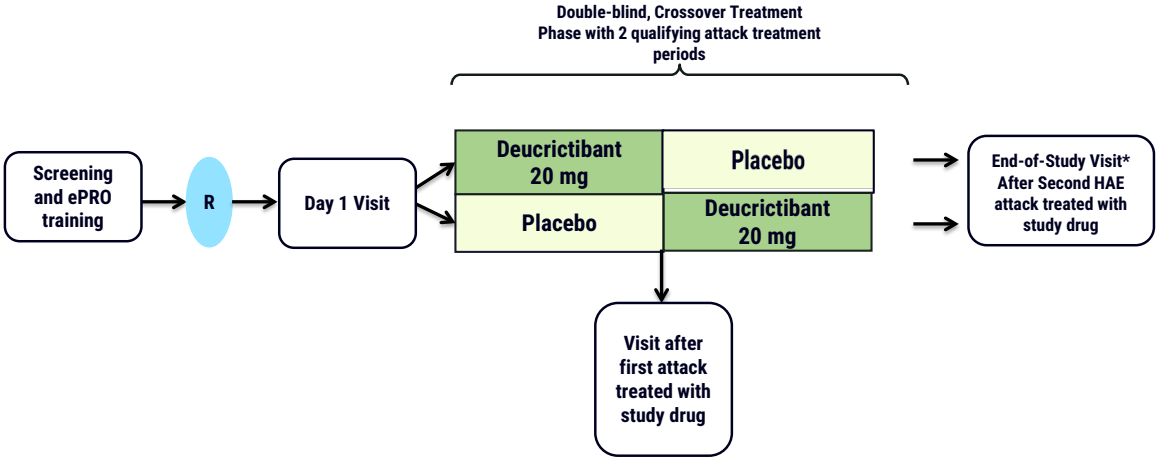
If the Sponsor elects to terminate the study, then the End-of-Study Visit for all remaining participants is to be completed within 30 days from the date that decision was communicated. (The study will officially close after all participants have completed the End-of-Study Visit.)

Qualitative interviews regarding experience with HAE standard-of-care medication (including double-blinded study drug), treatment preferences, and non-localized symptoms the participant typically experiences with HAE attacks (such as anxiety or fatigue), impairment of daily activities, as well as health-related quality of life as measured by the EuroQol 5 Dimension 5 Level (EQ-5D-5L), will be conducted ≥48 hours to ≤10 days following each of the 2 attacks treated with study drug, if allowed by local regulations. Interviews will be conducted by an external vendor via remote call. The qualitative interview following the second treated attack must take place ≥48 hours to ≤10 days following study drug administration and **before** completion of the End-of-Study Visit.

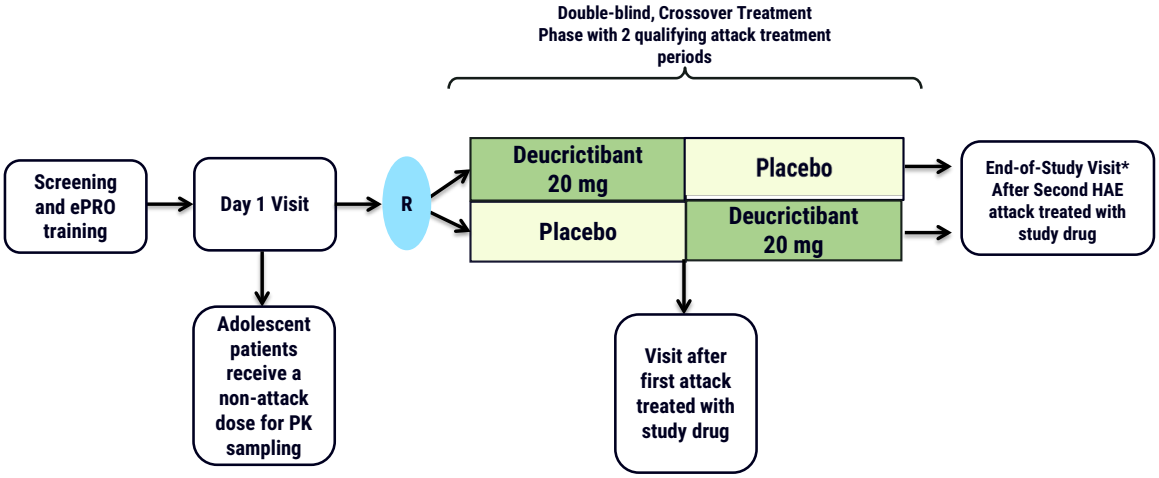
Participants who complete the current study will have an opportunity for continued access to deucrictibant in an open-label extension study, if inclusion and exclusion criteria are met.

Study Design Schematic:

Adults:



Adolescents:



Abbreviations: ePRO = electronic patient-reported outcome; HAE = hereditary angioedema; PK = pharmacokinetics; R = randomization.
*Data from the End-of-Study visit may be used to qualify the participant for an open-label extension study with deucrictibant.
*If the participant discontinues from the study prematurely, an End-of-Study visit will be performed within 10 (±5) days after the decision to discontinue study is communicated to the investigator. Participants who withdraw consent for further participation will be asked to attend an End-of-Study visit as soon as possible for safety assessments. If the Sponsor elects to terminate the study, then the End-of-Study visit for all remaining participants should be completed within 30 days from the date that decision was communicated. (The study will officially close after all participants have completed the End-of-Study Visit.)

Study Population:

The study will enroll approximately 120 adult and adolescent participants (≥12 to ≤75 years of age) with HAE. The study will target enrollment of approximately 9 adolescent participants (≥12 to <18 years of age at the time of screening).

The study is planned to be conducted at study sites in approximately 30 countries.

Inclusion Criteria:

Participants must meet all of the following criteria to be enrolled in the study:

1. Provision of written informed consent. If the participant is a minor (i.e., <18 years of age or as determined by local law), consent will be obtained from the participant's parent/legally designated representative/guardian, and written assent will be obtained from the participant, per country regulations.
2. Male or female, aged ≥12 to ≤75 years at the time of providing written informed consent/assent.
3. Diagnosis of HAE based upon the following:
   a. For participants with HAE type 1 or type 2 (type 1/2):
   - Documented clinical history consistent with HAE (subcutaneous or mucosal, nonpruritic swelling without accompanying urticaria)
   - At least one of the following:
     – Age ≤30 years at reported onset of first angioedema symptoms

- Family history consistent with HAE type 1/2
- C1q performed by the central laboratory is above the lower limit of the normal range

- Diagnostic testing results to confirm HAE type 1/2:
  - Participants on long-term prophylactic HAE therapies with C1-INH or danazol at the time of study entry must provide documentation of a historical laboratory test showing C1-INH functional level <50%.

    Note: If a confirmatory C1-INH test is required, the sample should be collected at least 5 half-lives after the last dose of C1-INH or danazol. This procedure will not be considered an interruption of stable LTP treatment.

    For all other participants with HAE type 1/2, C1-INH functional level of <50% must be shown by chromogenic assay performed by the central laboratory as part of the screening procedures.

b. For participants with HAE type 3:

- Recurrent angioedema attacks with diagnostic testing results obtained during screening to confirm C1-INH function ≥50% of normal and C4 level not below lower level of normal range performed by the central laboratory

  Note: The C1-INH sample should be collected at least 5 half-lives after the last dose of C1-INH therapy, if applicable.

- Must meet one of the following:
  - Documented genetic mutation associated with HAE type 3 as listed in the Hereditary Angioedema Association (HAEA) and World Allergy Organization (WAO)/European Academy of Allergy and Clinical Immunology (EAACI) Guidelines (Appendix 1)

  OR

  - If no documented mutation:
    - Clinical diagnosis with family history of HAE type 3 and an elevated BK peptide level confirmed by a commercially available assay

    AND

    - Attacks not responding to treatments with high-dose antihistamine (cetirizine 40 mg/day or equivalent high-dose second-generation antihistamine medication) and no clinical attack symptoms relief if treated with corticosteroid, montelukast, or omalizumab
- Documented effective attack symptom relief with on-demand icatibant treatment

4. History of at least 2 HAE attacks in the last 3 months before screening.
5. Experience with using standard-of-care treatment to effectively manage on-demand treatment for HAE attacks.
6. Either:

- Is on a stable dose and regimen of long-term prophylactic therapy with one of the following medications indicated for HAE: plasma-derived C1-INH, danazol at ≤200 mg/day, anti-fibrinolytics, berotralstat, or lanadelumab, for at least 6 months before the Screening Visit and intend to remain on the same dose for the duration of the study

OR

- Is receiving only on-demand treatment, and has not used long-term prophylactic treatment within the following durations before the Screening Visit, as specified below:
  - Plasma-derived C1-INH, berotralstat, anti-fibrinolytics, or attenuated androgens for at least 2 weeks
  - Lanadelumab for at least 10 weeks

7. Capable of recording, without assistance, electronic HAE diary and ePRO data using an electronic device, as evidenced by the competency assessment conducted during the Screening Phase.
8. For adolescent participants aged ≥12 to <18 years: body weight ≥40 kg.
9. Female participants of childbearing potential (or who become of childbearing potential during the study) must agree to the protocol specified pregnancy testing and to be abstinent from heterosexual intercourse or to use an acceptable contraception method, as defined in the protocol and as available locally, from enrollment until 30 days after the last study drug administration.

   There are no contraceptive requirements for male participants.
10. In the opinion of the Investigator, the participant (and parent/caregiver for adolescent participants) is willing and able to comply with the protocol.

Exclusion Criteria:

Participants who meet any of the following criteria will be excluded from the study.

1. Any female who is pregnant, plans to become pregnant, or is breastfeeding.
2. Any diagnosis of angioedema other than HAE.
3. Any clinically significant comorbidity or systemic dysfunction (e.g., cardiovascular, gastrointestinal, renal, neurological, respiratory) that, in the opinion of the Investigator, would interfere with the participant's safety or ability to participate in the study.
4. Use of attenuated androgens for short-term prophylaxis within 2 weeks before screening.
5. Abnormal hepatic function (aspartate aminotransferase [AST] >2×upper limit of normal [ULN], alanine aminotransferase [ALT] >2×ULN, or total bilirubin >1.5×ULN, or Child-Pugh class B or C). Participants with Gilbert's syndrome, being defined as an isolated increase in total bilirubin ≤3×ULN, with AST and ALT in the normal range, will not be excluded.
6. Abnormal renal function (estimated glomerular filtration rate <60 mL/min/1.73 m^2).
7. History of alcohol or drug abuse within the previous year, or current evidence of substance dependence or abuse.
8. Has received prior on-demand HAE treatment with deucrictibant.
9. Currently participating in any other investigational drug study or receiving other investigational treatment within the last 30 days, or within 5 half-lives (whichever is longer) of the time of randomization.
10. Prior gene therapy for any indication at any time.
11. Use of concomitant medications with systemic absorption that are strong inhibitors of CYP3A4 (such as clarithromycin, itraconazole, ketoconazole, and ritonavir) or strong inducers of CYP3A4 (such as carbamazepine and phenytoin) within the last 30 days or within 5 half-lives (whichever is longer) of the time of randomization.
12. Known hypersensitivity to deucrictibant or any of the excipients of study drug.

Test therapy, dosage, and mode of administration:

Deucrictibant is the active ingredient, previously referred to as PHA121 or PHA-022121.

The study drug consists of deucrictibant (20 mg strength) and matching placebo for oral use.

A study drug-treated attack refers to an attack that has been treated with orally administered deucrictibant 20 mg or matching placebo in a double-blinded fashion.

Timing of Treatment:

Day 1 Visit

Adolescents only (≥12 to <18 years) will receive deucrictibant (20 mg) in a non-attack state for PK assessment on Day 1 at the study site.

Treatment Phase

Adults and adolescents: Double-blinded study drug will be dispensed to each participant according to their randomized crossover treatment sequence for use during the Treatment Phase. Participants will self-administer oral doses of study drug: deucrictibant for the first qualifying attack followed by placebo for the second qualifying attack or the reverse order.

For study drug treated **non-laryngeal attacks**, a second dose of study drug is permitted ≥4 hours post first dose in case symptoms are persisting or progressing after the first study drug dose. If at ≥1 hour after the second dose, symptoms still persist or progress, then HAE on-demand rescue medication may be administered. Note that, if needed for participant safety, participants may self-administer HAE rescue medication at any time. The time of administration of study drug as well as HAE rescue medication must be recorded in the electronic device.

For study drug-treated **non-severe laryngeal attacks (in adults only)**, the participant may self-administer rescue medication at any time, if needed. A second dose of study drug should not be administered for non-severe laryngeal attacks.

The study drug should be taken with water and can be taken with or without meals. The timing of the most recent food intake before and after each study drug dose will be documented in the electronic HAE diary.

Rescue therapy, dosage, and mode of administration:

The participant can use locally approved and available on-demand HAE standard-of-care medication as rescue medication for treatment of qualifying HAE attacks if the study drug does not provide sufficient symptom relief as described in Section 6.1.2.3.1. Where feasible, commercially available icatibant will be provided as rescue medication to be used if needed.

Criteria for evaluation:

Efficacy:

Efficacy will be assessed with the following Patient Reported Outcomes (PROs):

- PGI-C (7-point scale) is used to evaluate the change in the HAE attack symptoms as compared to pre-treatment.
- PGI-S (5-point scale) is used to evaluate the severity of HAE attack symptoms.
- AMRA is used to evaluate the severity of HAE attack symptoms.
  - For non-laryngeal attacks, AMRA-3 is used for participants to rate from ‘0’ (No swelling/No pain) to ‘100’ (Worst possible swelling/Worst possible pain) for the severity of angioedema symptoms during an HAE attack. For abdominal or peripheral attacks, the AMRA-3 composite score is calculated as the mean of the single item scores for each of the 3 symptoms: skin pain, skin swelling, and abdominal pain.
  - For non-severe laryngeal attacks, the AMRA-5 is used for participants to rate the 5 major HAE symptoms between ‘0’ (No symptom) and ‘100’ (Worst symptom) for the severity of each of 5 angioedema symptoms during an HAE attack. The AMRA-5 score is calculated as the mean of the single item scores for each of the 5 symptoms: skin pain, skin swelling, abdominal pain, voice change, and difficulty swallowing.
- An interview to generate evidence on the content validity of AMRA-3/AMRA-5 in a subset of participants.

Safety:

- AEs
- Laboratory testing (blood chemistry, hematology, urinalysis, and pregnancy)
- Physical examination

- Vital signs (heart rate, blood pressure, body temperature)
- 12-lead ECG (digitally captured in triplicate)

PK (Adolescents Only):

- Blood samples for determination of deucrictibant plasma concentration-time profiles and PK parameters in adolescent participants

Exploratory (QoL):

- Qualitative interviews regarding experience with HAE medications (including double-blinded study drug), treatment preferences, non-localized symptoms the participant typically experiences with HAE attacks (such as anxiety or fatigue), impairment of daily activities, as well as health-related quality of life as measured by EQ-5D-5L via remote call, if allowed by local regulations

Statistical Methods:

The general information of study participants and baseline characteristics will be summarized based on all enrolled participants. All efficacy endpoints will be evaluated using the Primary Efficacy Analysis Set (PEAS, which consists of all randomized participants who had an attack treated with study drug in each of the two attack periods), unless otherwise specified. Supportive analyses will be performed using the Supportive Efficacy Analysis Set (SEAS, defined as all randomized participants who had at least one study drug treated attack) for selected endpoints, which will be detailed in the full Statistical Analysis Plan (SAP). Safety analysis will be based on the Safety Analysis Set.

For the primary endpoint, the estimated median time to onset of symptom relief, defined as PGI-C rating of at least "a little better" for 2 consecutive timepoints within 12 hours post-treatment, will be summarized for each treatment group using Kaplan-Meier (KM) method.

The intercurrent event (ICE) is the intake of rescue medication or prohibited medication after the attack is treated with study drug. Composite variable strategy will be used to address the ICE for the primary endpoint. The occurrence of ICE will be regarded as a treatment failure. Namely, if rescue medication or prohibited medication was taken within 12 hours post-treatment, the time to onset of symptom relief will be censored at the maximum time of consideration (i.e., 13 h, which is the upper end of the 12-hour time window in the electronic device for participant to record data at that nominal timepoint), regardless whether PGI-C reached the rating of at least "a little better".

For the primary endpoint, [redacted] will be used to test the null hypotheses that there is no difference between the deucrictibant and placebo treatments. Supportive analysis will be performed on SEAS using marginal Cox proportional hazards model (CPHM) with a robust variance-covariance estimator to account for the within-participant correlation. The model will include treatment (deucrictibant 20 mg or placebo), attack (attack 1 or attack 2), sequence (deucrictibant 20 mg/placebo or placebo/deucrictibant 20 mg), and randomization stratification factors as independent variables. The estimated hazard ratio between deucrictibant 20 mg and placebo will be provided with 95% confidence interval (CI) and p-value.

As supplementary analyses, the above analyses will be performed using a treatment policy strategy, in which the data from all efficacy assessments through 12 hours post-treatment, including those collected after intake of rescue medication or prohibited medication, will be used for the analysis of the primary endpoint.

For all secondary efficacy endpoints including time-to-event endpoints and binary endpoints, estimands will be defined in a similar way as for the primary endpoint, composite variable strategy will be consistently used to handle ICEs in all secondary efficacy endpoints.

The SAP will be signed off prior to database lock, and it will include a more technical and detailed description of all planned statistical analyses of all study endpoints.

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3. LIST OF ABBREVIATIONS AND DEFINITIONS OF TERMS

The following abbreviations and specialist terms are used in this study protocol.

Table 2: Abbreviations and Specialist Terms

Abbreviation	Definition
ADL	activities of daily living
AE	adverse event
ALT	alanine aminotransferase
AMRA-3	Angioedema SyMptom Rating scAle-3
AMRA-5	Angioedema SyMptom Rating ScAle-5
AST	aspartate aminotransferase
BK	bradykinin
C1-INH	C1 esterase inhibitor
CI	confidence interval
CPHM	Cox proportional hazards model
CRO	contract research organization
CTCAE	Common Terminology Criteria for Adverse Events
EAACI	European Academy of Allergy and Clinical Immunology
ECG	electrocardiogram
eCRF	electronic case report form
EDC	electronic data capture
EFD	embryofetal development
EoP	End of Progression
ePRO	electronic patient-reported outcome
EQ-5D-5L	EuroQol 5 Dimension 5 Level
FDA	Food and Drug Administration
FSH	follicle stimulating hormone
GCP	Good Clinical Practice
GEE	generalized estimating equation
HAE	hereditary angioedema

Abbreviation	Definition
HAE type 1/2	HAE type 1 or type 2
HAEA	Hereditary Angioedema Association
ICE	intercurrent event
ICF	informed consent form
ICH	International Council for Harmonisation
IDMC	Independent Data Monitoring Committee
IEC	Independent Ethics Committee
IND	Investigational New Drug
IRB	Institutional Review Board
IRT	interactive response technology
iv	intravenous
KM	Kaplan-Meier
MedDRA	Medical Dictionary for Regulatory Activities
NCI	National Cancer Institute
PEAS	Primary Efficacy Analysis Set
PGI-C	Patient Global Impression of Change
PGI-S	Patient Global Impression of Severity
PK	pharmacokinetics
Q	Quartile
QA	quality assurance
QoL	quality of life
SAE	serious adverse event
SAP	statistical analysis plan
sc	subcutaneous
SD	standard deviation
SEAS	Supportive Efficacy Analysis Set
SOC	system organ class
SOP	standard operating procedure

Abbreviation	Definition
SUSAR	Suspected Unexpected Serious Adverse Reaction
TEAE	treatment-emergent adverse event
ULN	upper limit of normal
VAS-3	visual analogue scale-3
VAS-5	visual analogue scale-5
WAO	World Allergy Organization

4. INTRODUCTION

4.1. Hereditary Angioedema

Hereditary angioedema (HAE) is a rare disease that manifests as unpredictable and recurrent edema attacks of subcutaneous or submucosal tissue most commonly affecting the limbs, intestinal tract, face (lips and tongue), genitals, and airways. HAE attacks often lead to discomfort and pain, and can include nausea, vomiting and diarrhea when the intestinal tract is affected. HAE attacks can become life-threatening in the case of airway obstruction and asphyxiation if the attack remains untreated. The number, duration, and severity of attacks vary highly between patients and throughout a patient's lifetime. Trauma, stress, and medical procedures are known triggers for attacks.

Three types of HAE have been characterized. HAE type 1 and type 2 (HAE type 1/2) are defined by mutations in the C1-INH gene and result either in reduced levels (type 1) or reduced activity (type 2) of C1-INH. In both cases, decreased C1-INH protease activity leads to uncontrolled activity of Factor XII and plasma kallikrein, resulting in an activated kallikrein-kinin system, i.e., excessive cleavage of high molecular weight kininogen and increased plasma levels of bradykinin (BK). HAE with normal C1-INH (HAE type 3) is characterized by similar clinical manifestations to HAE type 1/2 in the presence of normal levels and activity of C1-INH and, in a number of cases, is associated with characterized mutations in other genes as listed in the HAEA and WAO/EAACI Guidelines (Appendix 1). Laboratory tests that measure bradykinin levels in blood and stimulated kallikrein activity are available to aid identification of patients with HAE type 3 (Li 2023; Lara-Marquez 2018).

BK is the principal mediator of the signs and symptoms associated with HAE attacks. Two BK receptor types, bradykinin B1 and bradykinin B2, have been identified, cloned, and characterized. BK has a high affinity and high agonist potency at the bradykinin B2 receptor, but a near 1000-fold lower affinity and potency at the B1 receptor, making it a selective bradykinin B2 receptor agonist.

The bradykinin B2 receptor is expressed on the cell surface of the luminal side of vascular endothelial cells. Excessive BK generation, as in HAE, promotes vascular permeability, leading to plasma extravasation and subcutaneous or submucosal tissue edema and swelling typical of an angioedema attack.

4.2. Current Treatment Options for Hereditary Angioedema

There are currently 2 treatment approaches to the management of HAE: acute (on-demand) treatment of attacks and prevention of attacks with short- or long-term prophylactic therapy. Patients and their physicians have several choices for on-demand or prophylactic treatment, but because there are differences in the route of administration, side effect profiles, and efficacy between medications, patient preference and experience are considered in selecting the most appropriate therapy.

Patients must have access to a readily available on-demand HAE standard-of-care medication that can be self-administered. Treatment should be taken regardless of the location of the attack, and treatment should be initiated early to mitigate progression of the attack. Regular monitoring of the patient's frequency of use of on-demand treatment and treatment outcomes should be

conducted (Busse 2021, Maurer 2022). In addition to on-demand treatment of HAE attacks, a patient may also require prophylactic treatment.

The requirement for long-term prophylactic treatment is not made on rigid criteria, but rather the needs of the individual patient (Busse 2021, Maurer 2022). The decision to initiate long-term prophylaxis is made with consideration for the patient's quality of life and treatment preferences in the context of attack frequency, attack severity, comorbid conditions, and access to treatment (Busse 2021, Maurer 2022). Because disease severity may change over time, the need to initiate or continue long-term prophylactic treatment is periodically reviewed.

4.2.1. On-demand Treatments

The bradykinin B2 receptor antagonist icatibant (Firazyr®) (Firazyr® prescribing information, 2021) is the mainstay of acute treatment due to its quick onset of action and subcutaneous (sc) self-administration with prefilled syringe. The kallikrein inhibitor ecallantide (Kalbitor®) requires hospital or health professional sc administration due to potential anaphylaxis. Plasma-derived (Berinert® 500/1,500, Cinryze®) or recombinant (Ruconest®) C1-INH concentrates require intravenous (iv) administration.

4.2.2. Prophylactic Treatments

The plasma-derived C1-INH concentrates (Cinryze and Haegarda®/Berinert® 2,000/3,000) are administered by iv and sc routes, respectively. These products are administered twice weekly (every 3 or 4 days). Plasma kallikrein inhibitors include lanadelumab (Takhzyro®), a fully human recombinant monoclonal antibody, which is administered sc every 2 or 4 weeks, and berotralstat (Orladeyo®), which is administered orally, once daily. Attenuated androgens (including danazol, stanozolol, oxandrolone, oxymetholone, tibolone, methyltestosterone) and antifibrinolytics, such as ε-aminocaproic acid and tranexamic acid, are used for the prevention of acute attacks and are licensed in some countries. However, current treatment guidelines recommend against the use of attenuated androgens and antifibrinolytics as first-line therapy due to limited efficacy and/or safety issues with associated contraindications (Busse 2021, Maurer 2022).

4.3. Unmet Medical Needs in Managing Bradykinin-Mediated Angioedema

There is a high unmet medical need for efficacious oral medicines that can facilitate timely treatment of attacks due to HAE and other various forms of bradykinin-mediated angioedema and thereby improve outcomes, compliance, and the overall care and quality of life for patients. All currently available approved medications for on-demand treatment of HAE are administered parenterally via iv or sc routes, which are often associated with injection pain, local side effects, a delay of drug administration, and a lower quality of life. In a 2017 Food and Drug Administration (FDA) Patient-Focused Drug Development meeting, patients expressed that the route of administration (iv, sc, or oral) is the most important consideration for a new HAE therapeutic, with a preference for oral treatments (FDA Report of Public Meeting 2018).

4.4. Deucrictibant

Deucrictibant is under development as an orally administered BK B2 receptor antagonist with potential for both acute therapeutic and prophylactic use in patients with different types of BK-

mediated angioedema, including HAE type 1/2, HAE with normal C1-INH (HAE type 3), and acquired angioedema due to C1 deficiency.

Deucrictibant has been extensively characterized in primary pharmacological studies as a potent, orally available B2 receptor antagonist.

Deucrictibant had no adverse effects on male or female fertility or embryofetal development (EFD) in the humanized bradykinin B2-receptor transgenic rat nor on EFD in rabbits, up to the highest dose tested. This supports the inclusion of women of childbearing potential in the Phase 3 clinical studies provided appropriate contraception precautions are in place. Functional assessment of male fertility has been completed both in the wild-type rat as part of the 6-month repeat-dose toxicity study and in a dedicated male fertility study in the humanized bradykinin B2-receptor transgenic rat following pre-mating dosing of at least 10 weeks. There are no effects on male reproductive parameters up to the highest dose tested. In addition, there were no organ weight or histopathological changes of concern in the rat reproductive tract of either sex. In conjunction with the EFD results, the clinical studies do not require male participants to use male contraception.

Overall, in completed Phase 1 studies (using either deucrictibant oral solution or oral soft capsule), in the Phase 2 clinical study of on-demand deucrictibant in adults with HAE (PHA022121-C201), and in the ongoing 2-part extension study of on-demand deucrictibant in adults with HAE (PHA022121-C303), deucrictibant has been shown to be generally well tolerated. Most treatment-emergent AEs reported have been mild to moderate in severity. No severe treatment-emergent adverse events (TEAEs) or serious adverse events (SAEs) have been considered by the investigator as related to deucrictibant. No serious adverse reactions or deaths have been reported.

In Study PHA022121-C201, single-dose PK was characterized and participants treated up to 3 qualifying HAE attacks with blinded study drug. Efficacy data from the completed primary analysis of the HAE attack period showed that deucrictibant 10, 20, and 30 mg, versus placebo, had a significantly shorter time to onset of symptom relief by a ≥30% reduction in 3-symptom composite visual analogue scale (VAS-3) score from the pre-treatment score, shorter time to a ≥50% reduction in VAS-3 score from the pre-treatment score, shorter time to almost complete (i.e., VAS <10) or complete (VAS=0) symptom relief per VAS-3, greater decrease in mean symptom complex severity score from pre-treatment to 4 hours post-treatment, and improved treatment outcome score at 4 hours post-treatment. Deucrictibant was well-tolerated at all 3 dose levels. All TEAEs within 48 hours following administration of deucrictibant were Grade 1 or 2 (mild or moderate) in intensity, and the following were assessed by the investigator as treatment-related in 1 participant each (headache, nausea, vomiting, and fatigue). No treatment-related severe TEAEs or SAEs were reported. There were also no TEAEs leading to treatment or study discontinuation nor any treatment-related TEAEs related to abnormal laboratory parameters, vital signs, or electrocardiogram (ECG) parameters. These data support further development of deucrictibant as an on-demand treatment for patients with HAE. The results of the full analysis of Study PHA022121-C201 support those of the primary analysis.

The 3-symptom composite visual analogue scale (VAS-3; Lumry 2011, Kusuma 2012) has been renamed by the sponsor as the 3-symptom composite Angioedema SyMptom Rating scAle (AMRA-3) for assessment of non-laryngeal attacks and is henceforth referred to as AMRA-3 in this protocol. Similarly, the 5-symptom composite visual analogue scale (VAS-5) has been

renamed by the sponsor as the 5-symptom composite Angioedema SyMptom Rating scAle-3 (AMRA-5) for assessment of non-severe laryngeal attacks and is henceforth referred to as AMRA-5 in this protocol. Collectively, the tool is referred to as AMRA.

4.4.1. Benefit-Risk with Deucrictibant

Overall, deucrictibant has been generally well-tolerated in the completed and ongoing clinical studies. The majority of TEAEs reported as of the safety data cutoff date have been mild/Grade 1 or moderate/Grade 2 in severity and have resolved. TEAEs in assigned deucrictibant treatment group reported by ≥2 participants with frequency ≥5% from studies completed or with predefined primary data analysis have included headache, dizziness, nasopharyngitis, gamma glutamyl transferase increased, urinary tract infection, and diarrhea. Similar events were also reported in participants receiving placebo. No serious adverse reactions (i.e., study drug-related SAEs) or TEAEs with an outcome of death have been identified thus far. Independent Data Monitoring Committees have been established to review the unblinded safety data from all of the studies on a regular basis and ensure the safety of participants. Safety monitoring rules have been defined in study protocols to mitigate the risks for participants in clinical studies of deucrictibant.

In the Phase 2 double-blind, placebo controlled and randomized on-demand Study PHA022121-C201 in patients with HAE, deucrictibant demonstrated robust efficacy measured by rapid onset of attack symptom relief, substantial reduction of attack symptoms, early attack symptom resolution, and markedly reduced use of rescue medication compared to placebo.

Given the potential benefit to treat the HAE population, especially with the validated mechanism of action, positive pharmacodynamic and clinical effects demonstrated by the BK challenge in Study PHA022121-C001 and on-demand treatment of HAE attacks in Study PHA022121-C201, the benefit-risk assessment of deucrictibant is considered acceptable if used according to the clinical study protocol, and in accordance with the recommendations and guidance provided in the Investigator's Brochure.

Overall, the benefit-risk assessment appears to be favorable to support the clinical investigation of deucrictibant for on-demand treatment of HAE.

Refer to the Investigator's Brochure for details regarding the benefit-risk with deucrictibant.

5. STUDY OBJECTIVES AND ENDPOINTS

The study objectives and endpoints are listed in Table 3.

Table 3: Objectives and Endpoints

Objectives	Endpoints
Primary	
• To evaluate the efficacy of deucrictibant as on-demand treatment compared with placebo on the onset of symptom relief during HAE attacks	• Time to onset of symptom relief, defined as Patient Global Impression of Change (PGI-C) rating of at least "a little better" for 2 consecutive timepoints within 12 hours post-treatment
Secondary (Efficacy)	
• To evaluate the efficacy of deucrictibant as on-demand treatment compared with placebo on symptom relief and resolution of HAE attacks	1. Proportion of study drug-treated attacks achieving PGI-C rating of at least "a little better" at 4 hours post-treatment 2. Time to substantial symptom relief, defined as achieving PGI-C rating of at least "better" for 2 consecutive timepoints within 12 hours post-treatment 3. Time to substantial symptom relief by Patient Global Impression of Severity (PGI-S), defined as achieving ≥1 point reduction in PGI-S from pre-treatment for 2 consecutive timepoints within 12 hours post-treatment 4. Time to complete symptom resolution, defined as achieving PGI-S rating of "none" within 48 hours post-treatment 5. Time to End of Progression (EoP) in attack symptoms within 12 hours, with EoP time defined as the earliest post-treatment timepoint after which all subsequent PGI-C ratings are stable or improved 6. Proportion of study drug-treated attacks requiring rescue medication within 24 hours post-treatment 7. Proportion of attacks achieving symptom resolution, defined as achieving PGI-S rating of "none" with one dose of study drug at 24 hours post-treatment

Objectives	Endpoints
	8. Time to substantial symptom relief by Angioedema SyMptom Rating scAle (AMRA), defined as a ≥50% reduction in AMRA composite score from pre-treatment for 2 consecutive timepoints within 12 hours post-treatment 9. Time to almost complete or complete symptom relief by AMRA, defined as all item scores in AMRA having a value ≤10 for 2 consecutive timepoints within 24 hours post-treatment 10. Proportion of study drug-treated attacks reaching almost complete or complete symptom relief by AMRA, defined as all item scores in AMRA having a value ≤10 at 24 hours post-treatment 11. Time to EoP in attack symptoms within 12 hours, with EoP time defined as the earliest post-treatment timepoint after which every individual AMRA item is stable or improved at all subsequent timepoints
Secondary (Safety)	
To evaluate the safety and tolerability of deucrictibant compared with placebo for on-demand treatment of HAE attacks	Safety endpoints: • Incidence of TEAEs and SAEs • Change from baseline in clinical laboratory, vital sign, and ECG parameters
Secondary (Pharmacokinetics [PK]) – Adolescents Only	
To assess the single-dose pharmacokinetics (PK) of deucrictibant in adolescent participants (≥12 to <18 years) in a non-attack state	PK endpoints: • Deucrictibant plasma concentration-time profiles • PK parameters
Exploratory:	
To evaluate participants' quality of life	Exploratory endpoint: • Changes in health-related quality of life (QoL)

6. INVESTIGATIONAL PLAN

6.1. Overall Study Design

This is a Phase 3, multicenter, randomized, double-blind, placebo-controlled, 2-period, 2-treatment cross-over study to evaluate the efficacy and safety of orally administered deucrictibant for the on-demand treatment of attacks in participants ≥12 to ≤75 years of age with HAE (type 1, type 2, or type 3), a proportion of whom are using long-term prophylactic medication for HAE.

The study consists of a Screening Phase during which eligibility is confirmed, a Treatment Phase in which participants will be randomized and receive double-blinded study drug to treat 2 qualified HAE attacks (i.e., 2 Treatment Periods within the Treatment Phase) at home, and an End-of-Study Follow-up Phase after the second study drug-treated attack. In addition, for adolescent participants (age ≥12 to <18 years), PK samples are collected after administration of deucrictibant at Day 1 in a non-attack state.

After signing the informed consent form (ICF)/assent form, participants will be screened for eligibility. During the Screening Phase, participants will attend an on-site visit to confirm eligibility, to undergo screening procedures, and to receive training on the use of the electronic device with the electronic patient-reported outcome (ePRO) instruments and the electronic HAE diary. The participants will complete an ePRO competency assessment (attack simulation) to ensure their ability to complete required ePRO assessments and questionnaires with the electronic device during an actual attack during the Treatment Phase.

After confirmation of eligibility, participants will be randomized on Day 1 to a crossover treatment sequence of double-blinded study drug in a 1:1 ratio (deucrictibant 20 mg for the first attack, placebo for the second attack or the reverse order) to treat 2 qualifying attacks, with each attack could be either a qualifying non-laryngeal attack or non-severe laryngeal attack not associated with breathing difficulties or stridor, during the Treatment Phase.

Randomization will be stratified according to participant age (≥12 to <18 years, ≥18 years) and use of long-term prophylactic HAE therapy (Yes/No).

Refer to Section 6.1.2.3.1 for details regarding attack qualification by the investigator (or designee) and treatment, based on attack type.

After the participant self-administers study drug for the first qualifying HAE attack, they will have either an on-site or remote Post Attack Visit ≥48 hours to ≤10 days after study drug administration for evaluation and recording of AEs and concomitant medication use.

For study drug treated **non-laryngeal attacks**, a second dose of study drug is permitted ≥4 hours post first dose in case symptoms are persisting or progressing after the first study drug dose. If at ≥1 hour after the second dose, symptoms still persist or progress, then HAE on-demand rescue medication may be administered. Note that, if needed for participant safety, participants may self-administer HAE rescue medication at any time. The time of administration of study drug as well as HAE rescue medication must be recorded in the electronic device.

For study drug-treated **non-severe laryngeal attacks (in adults only)**, the participant may self-administer rescue medication at any time, if needed. A second dose of study drug should not be administered for non-severe laryngeal attacks.

After the participant self-administers study drug for the second qualifying HAE attack, an on-site End-of-Study Visit is to occur at 10 (±5) days following attack resolution.

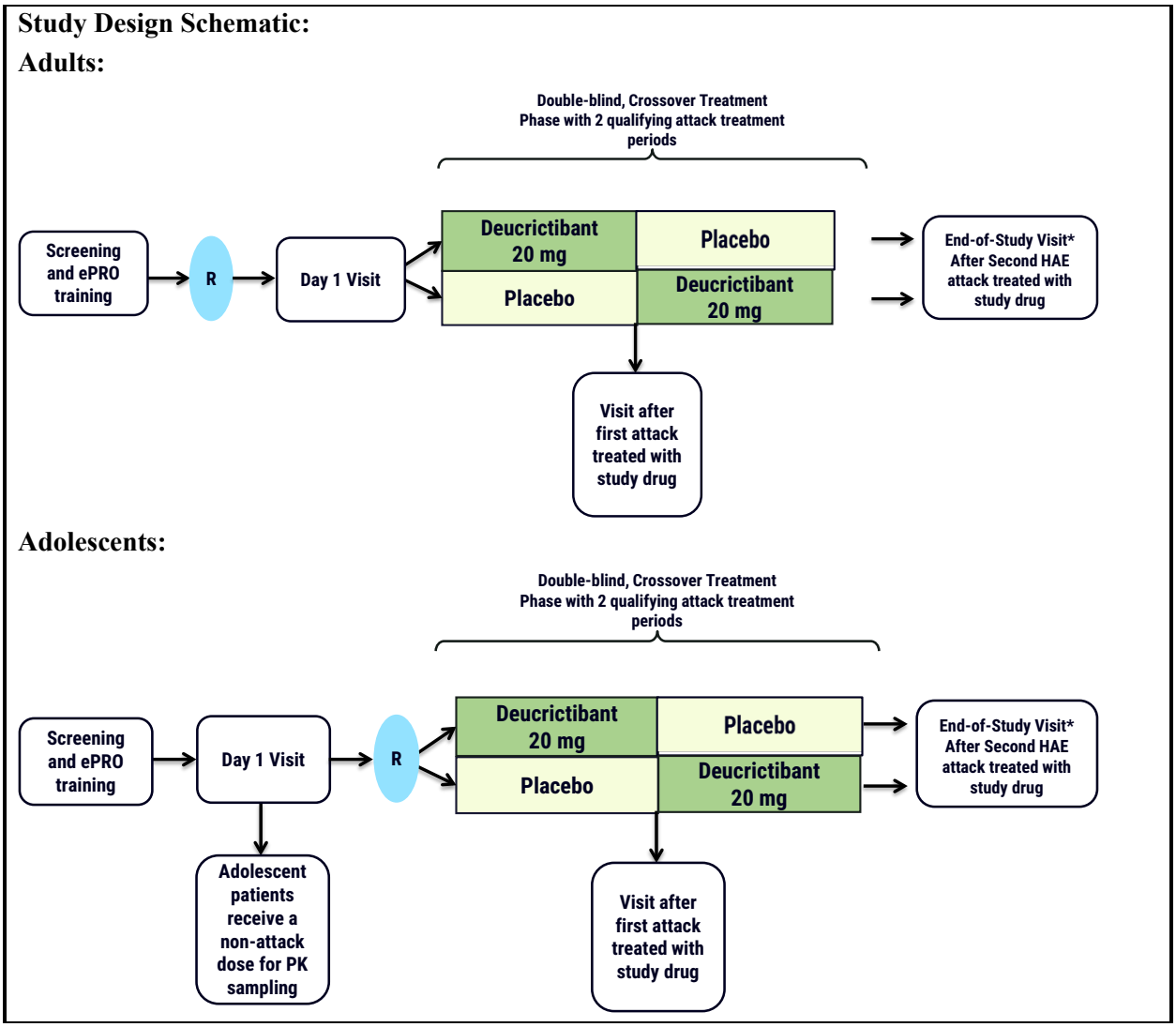
For participants who discontinue from the study prematurely, the End-of-Study Visit will be performed within 10 (±5) days after the decision to discontinue the study is communicated to the Investigator. Participants who withdraw consent for further participation will be asked to attend the End-of-Study Visit as soon as possible for safety assessments.

If the Sponsor elects to terminate the study, then the End-of-Study Visit for all remaining participants should be completed within 30 days from the date that decision was communicated. (The study will officially close after all participants have completed the End-of-Study Visit.)

Qualitative interviews regarding experience with HAE standard-of-care medication (including double-blinded study drug), treatment preferences, and non-localized symptoms the participant typically experiences with HAE attacks (such as anxiety or fatigue), impairment of daily activities, as well as health-related quality of life as measured by EQ-5D-5L, will be conducted ≥48 hours to ≤10 days following each of the 2 attacks treated with study drug, if allowed by local regulations. Interviews will be conducted by an external vendor via remote call. The qualitative interview following the second treated attack must take place ≥48 hours to ≤10 days following the study drug administration and **before** completion of the End-of-Study Visit.

Participants who complete the current study will have an opportunity for continued access to deucrictibant in an open-label extension study, if inclusion and exclusion criteria are met.

Figure 1 Study Design



Abbreviations: ePRO = electronic patient-reported outcome; HAE = hereditary angioedema; PK = pharmacokinetics; R = randomization.
*Data from the End-of-Study visit may be used to qualify the participant for an open-label extension study with deucrictibant.
*If the participant discontinues prematurely, an End-of-Study visit will be performed within 10 (±5) days after the decision to discontinue study is communicated to the investigator. Participants who withdraw consent for further participation will be asked to attend an End-of-Study visit as soon as possible for safety assessments. If the Sponsor elects to terminate the study, then the End-of-Study visit for all remaining participants should be completed within 30 days from the date that decision was communicated. (The study will officially close after all participants have completed the End-of-Study Visit.)

6.1.1. Schedule of Events

The Schedule of Events is presented in Table 4.

Table 4: Schedule of Events

	Screening Phase [a]	Day 1 Visit	Treatment Phase (First and Second Attack Period)	Post Attack Visit for 1st Attack	End-of-Study Visit
	Day -28 to -1	Day 1	Approx 36 weeks	≥48 hours to ≤10 days after 1st study drug taken	10 days (±5) after attack resolution [c]
	On-site	On-site/Remote [b]	Remote	On-site/Remote	On-site & Remote
Informed consent [d]	X				
Demographics and HAE history [e]	X				
Medical history	X				
Height (screening only) and weight	X				X
Physical examination	X				X
Vital signs [f]	X				X
12-lead ECG (digital triplicate) [g]	X				X
C1-INH, C1q, and C4 testing for eligibility (as applicable) [h]	X				
Safety laboratory testing [i]	X				X
FSH test [j]	X				
Pregnancy test [k]	X				X
Adverse events [l]	◄	———	———	———	►
Prior/concomitant medications	◄	———	———	———	►
Electronic HAE diary and ePRO training, issue electronic device, and competency assessment by participant	X			X[m]	X[m]
Randomization		X			
Dispense study drug for first qualifying attack		X [n]			
Dispense rescue medication		X		X (if necessary) [o]	
Dispense/ship study drug for second qualifying attack				X	

	Screening Phase [a]	Day 1 Visit	Treatment Phase (First and Second Attack Period)	Post Attack Visit for 1st Attack	End-of-Study Visit
	Day -28 to -1	Day 1	Approx 36 weeks	≥48 hours to ≤10 days after 1st study drug taken	10 days (±5) after attack resolution [c]
	On-site	On-site/Remote [b]	Remote	On-site/Remote	On-site & Remote
Adolescents only (≥12 to <18y): [p]					
Administer deucrictibant (20 mg)		X			
Collect PK plasma samples		X			
Participant to confirm with Investigator/designee that attack qualifies, then self-administer study drug			X [q]		
Electronic daily HAE diary [r]		◄———	———	———►	
Recording of meal intake [s]		X [s]			
ePRO: Body map [t]			X		
ePRO: PGI-C [u]			X		
ePRO: PGI-S [v]			X		
ePRO: AMRA [w]			X		
Qualitative interview (remote) [x]				X [x]	X
AMRA Interview (remote) [y]		X			
Return of electronic device					X

Abbreviations: EQ-5D-5L = EuroQol 5 Dimension 5 Level; AE = adverse event; AMRA-3 = Angioedema SyMptom Rating scAle; AMRA-5 = Angioedema SyMptom Rating scAle; ECG=electrocardiogram; FSH=follicle stimulating hormone; ePRO=electronic participant reported outcome (instrument); HAE=hereditary angioedema; hCG = human chorionic gonadotropin; PGI-C=participant global impression of change; PGI-S=participant global impression of severity, PK=pharmacokinetic.

a. Screening Phase may take up to 4 weeks from signing of informed consent until randomization.
b. For adult participants, the Day 1 visit may be performed remotely if study drug can be shipped to the participant, as allowed by local regulations.
c. After the participant self-administers study drug for the second qualifying HAE attack, an on-site End-of-Study Visit is to occur at 10 (±5) days following attack resolution. For participants who discontinue from the study prematurely, the End-of-Study Visit will be performed within 10 (±5) days after the decision to discontinue the study is communicated to the Investigator. Participants who withdraw consent for further participation will be asked to attend the End-of-Study Visit as soon as possible for safety assessments. If the Sponsor elects to terminate the study, then the End-of-Study Visit for all remaining participants should be completed within 30 days from the date that decision was communicated. (The study will officially close after all participants have

completed the End-of-Study Visit.) Data from the End-of-Study Visit may be used to qualify the participant for an open-label extension study of deucrictibant soft capsule.

d. Informed consent/assent must be signed by the participant and, if applicable, informed consent must be signed by the participant's parent/legally designated representative/guardian, before any study-specific procedures are performed. If an adolescent participant reaches the age of 18 years during study participation, the participant will be required to provide written informed consent in order to continue study participation, in accordance with local regulations.
e. After preliminary eligibility according to the inclusion and exclusion criteria has been established, the study site will submit required documentation supporting eligibility to the Sponsor or designee for review prior to enrollment.
f. Blood pressure, heart rate, and body temperature.
g. A digital 12-lead ECG (recorded in triplicate) will be performed and recorded, after the participant has rested in the supine position for at least 5 minutes at the Screening Visit, and at the End-of-Study Visit. Central reading of all recorded ECGs will be performed by the ECG vendor. ECGs will be evaluated by the Investigator based on the central reading. If the reading is abnormal, the Investigator will assess whether it is "Clinically significant" or "Not clinically significant", and recorded in the eCRF.
h. Collection of a plasma sampling for central laboratory assessment of C1-INH functional level (for participants not on long-term prophylaxis with C1-INH or danazol), and C1q level. Note: If a confirmatory C1-INH test is required, the sample should be collected at least 5 half-lives after the last dose of C1-INH or danazol. This procedure will not be considered an interruption of stable LTP treatment. For participants with HAE type 3 only: diagnostic testing results to confirm C1-INH function ≥50% of normal and C4 level not below lower level of normal range performed by the central laboratory.
i. Includes hematology, blood chemistry, coagulation testing, and macroscopic dipstick urinalysis. Microscopic examination is to be performed only if macroscopic findings are abnormal. Unscheduled clinical laboratory testing is permitted if deemed necessary for safety by the Investigator and/or Sponsor. Additional safety sampling may be performed and processed by a local laboratory at the same time points as the central laboratory, as per the Schedule of Events. Based on Investigator determination or local requirement for participant safety, additional samples may be assessed during the study by a local laboratory, in parallel with the central laboratory.
j. For female participants who are post-menopausal (defined as 12 consecutive months of amenorrhea without an alternative cause), a serum sampling to test for FSH levels should be performed at Screening to confirm post-menopausal status.
k. For female participants of childbearing potential, a urine hCG test must be performed. Serum pregnancy testing and/or more frequent pregnancy testing will be performed if required by local regulations or in case of suspected pregnancy.
l. All AEs with onset after signing informed consent/assent to the last follow-up at End-of Study Visit are to be recorded.
m. After each attack, study center personnel may retrain participants on use of the electronic device, at their discretion.
n. Following randomization, participants will be dispensed 2 doses of study drug for on-demand treatment of the first qualifying attack. If necessary due to logistical challenges, randomization may take place one day before the Day 1 visit; however, no study drug will be dispensed to the participant until Day 1.
o. If a remote visit is conducted for the Post Attack Visit, 2 additional doses of study drug may be shipped via courier from the study site to the participant after the Post Attack Visit has been completed and all study drug dispensed for the first attack has been returned (eg, via courier) to the site.
p. Adolescent participants will receive a single dose of oral deucrictibant (20 mg) to assess PK and safety (AEs). Plasma samples will be collected at pre-dose, and post-dose at 15 minutes (±5), 30 minutes (±5), 1 hour (±10 minutes), 2 hours (±10 minutes), 4 hours (±15 minutes), and 8 hours (±15 minutes).
q. Refer to Section 6.1.2.3.1 for attack qualification and treatment procedures.
r. Participants will complete an HAE diary via an electronic device to record daily whether they experience an attack and, if yes, record the start and stop date of any HAE attack. HAE attacks not meeting all the criteria in Section 6.1.2.3.1 should be treated with usual standard-of-care HAE medication. Participants should seek immediate medical attention for HAE attacks with breathing difficulties or stridor.
s. Each time study drug is administered, meal intake before and after study drug administration will be documented in the electronic device for all participants.
t. The body map is completed pre-treatment only.

u. PGI-C: at same time points as AMRA (except pre-treatment; see footnote w), including just before taking a second dose of study drug and/or rescue medication.
v. PGI-S: at same time points as AMRA (see footnote w), including just before taking a second dose of study drug and/or rescue medication.
w. AMRA score (AMRA-3 for non-laryngeal attacks/AMRA-5 for non-severe laryngeal attacks without breathing difficulties or stridor): pre-treatment at the time the Investigator is consulted to confirm qualification of the attack, at 15 and 30 minutes after self-administration of study drug, then every 30 minutes through 4 hours post-treatment, and at 5, 6, 7, 8, 9, 10, 11, 12, 14, 16, 18, 20, 22, 24, 36, and 48 (+6) hours post-dose. For any use of a second dose of study drug or rescue medication, the AMRA should be recorded on the electronic device just before taking the second dose and/or rescue medication.
x. Qualitative interviews regarding experience with HAE standard-of-care medication (including double-blinded study drug), treatment preferences, and non-localized symptoms the participant typically experiences with HAE attacks (such as anxiety or fatigue), impairment of daily activities, as well as health-related quality of life as measured by EQ-5D-5L, will be conducted ≥48 hours to ≤10 days following each of the 2 attacks treated with study drug, if allowed by local regulations. Interviews will be conducted by an external vendor via remote call. The qualitative interview following the second treated attack must take place ≥48 hours to ≤10 days following the study drug administration and before completion of the End-of-Study Visit.
y. In a subset of participants at selected sites, an interview to generate evidence on the content validity of AMRA-3/AMRA-5 will be conducted within 72 hours after the Day 1 Visit. The interview will be conducted by an external vendor via remote call.

6.1.2. Study Activities

All study procedures to be performed are described below by visit, with the Schedule of Events provided in Table 4.

6.1.2.1. Screening Phase (Days -28 to -1)

The Screening Phase may take up to 4 weeks. After signing the informed consent/assent, participants will be screened for eligibility according to the inclusion and exclusion criteria. The following assessments will be performed at the site:

- Demographics and HAE characteristics
- Medical history
- Weight and height
- Physical examination
- Vital signs
- Digital triplicate 12-lead ECG
- C1-INH, C1q, and C4 testing (as applicable). Note: If a confirmatory C1-INH test is required, the sample should be collected at least 5 half-lives after the last dose of C1-INH or danazol therapy. This procedure will not be considered an interruption of stable LTP treatment. For participants with HAE type 3 only: diagnostic testing results to confirm C1-INH function ≥50% of normal and C4 level not below lower level of normal range performed by the central laboratory.
- Clinical laboratory testing (hematology, biochemistry, coagulation, urinalysis)
- Urine pregnancy test (for women of childbearing potential)
- Follicle-stimulating hormone (FSH) testing (for women who have 12 consecutive months of amenorrhea without an alternative medical cause and are considered post-menopausal), using serum sampling
- Adverse events (reporting of AEs occurs from date of signed informed consent/assent through the End-of-Study Visit, inclusive; see Section 11.11 for details)
- Prior and/or concomitant medications
- Training on electronic ePRO instruments (body map, AMRA-3/5, PGI-S, and PGI-C) and the electronic HAE diary using an electronic device

In addition, participants will complete an ePRO competency assessment (attack simulation), with retraining and repeat assessment, at study center personnel discretion, until successful, to ensure their ability to complete required ePRO assessments and questionnaires during an actual attack during the treatment phase. Details will be provided in a separate study manual.

After the on-site Screening Visit, the Investigator will confirm whether the participant is eligible for the study after the results of all safety laboratory tests have become available and will submit documentation supporting eligibility to the Sponsor or designee for review prior to enrollment.

6.1.2.2. Day 1 Visit

After confirmation of eligibility, participants will be randomized to a treatment sequence of blinded study drug in a 1:1 ratio (deucrictibant [20 mg] for the first attack, placebo for the second attack, or the reverse order) to treat 2 qualifying attacks. Participants will be randomized during the Day 1 Visit. If necessary due to logistical challenges, randomization may take place one day before the Day 1 Visit, however, no study drug will be dispensed to the participant until Day 1. Randomization of eligible adolescent participants should preferably occur only after the non-attack dose is administered on Day 1, and it is determined that there are no safety concerns based on that dose.

For adult participants, the Day 1 visit may be performed remotely if study drug can be shipped to the participant, as allowed by local regulations. If a participant opts to have their Day 1 Visit performed remotely, randomization should take place one day before the Day 1 Visit, so that study drug can be shipped to the participant's home in time for the Day 1 Visit.

Other activities performed during the Day 1 Visit:

- Retraining on the electronic HAE diary, ePRO instruments, and other study requirements (e.g., AEs, concomitant medications) will be performed, at study center personnel discretion.
- Study drug (2 doses) and rescue medication will be dispensed for the participant to take home to treat the first qualifying HAE attack.
- AEs
- Concomitant medications
- Adolescents only (≥12 to <18 years): All adolescent participants (but not adult participants) will receive a single deucrictibant (20 mg) in a non-attack state to assess PK of deucrictibant and safety (AEs). Plasma samples will be collected pre-dose, and post dose at 15 minutes (±5), 30 minutes (±5), 1 hour (±10 minutes), 2 hours (±10 minutes), 4 hours (±15 minutes), and 8 hours (±15 minutes).
- In a subset of participants at selected sites and if allowed by local regulations, AMRA interview to support the development of the AMRA-3/AMRA-5 endpoint

6.1.2.3. Treatment Phase

After Day 1, the participant is to complete the electronic HAE diary on a daily basis. All occurrences of HAE attacks, regardless of whether treated with study drug, are to be documented in the electronic HAE diary throughout the study. The intake of study drug, second dose of study drug (if applicable), and rescue medication (if applicable) will be recorded in the electronic device. If a participant experiences an HAE attack any time between signing the informed consent and the Day 1 visit, the participant should treat the HAE attack with their usual standard-of- care HAE medication. For adolescents, the Day 1 Visit should be postponed until symptoms have resolved and at least 72 hours have elapsed since HAE standard-of-care medication was taken.

During the treatment phase, the participant will record the use of concomitant medications on a paper worksheet to be shared with the study site for entry into the electronic case report form (eCRF).

Two qualifying HAE attacks, either non-laryngeal or non-severe laryngeal (without breathing difficulties or stridor), are to be treated with study drug during the Treatment Phase. Thus, the Treatment Phase consists of 2 Treatment Periods, 1 each around each of the 2 qualified HAE attacks.

6.1.2.3.1. Attack Qualification and Treatment, based on Attack Type

The Investigator will confirm whether an attack is non-laryngeal or laryngeal in nature and whether it qualifies for study drug administration. The Investigator will be provided with a worksheet to facilitate the attack qualification assessment and required data collection. Non-severe laryngeal attacks (i.e., laryngeal attacks without breathing difficulties or stridor) eligible for study drug administration are characterized with symptoms associated with larynx and/or pharynx, such as sensation of lump in the throat, difficulty swallowing, and/or voice change. Treatment of qualifying non-severe laryngeal attacks with study drug will be permitted in adult participants only.

6.1.2.3.1.1. Non-Laryngeal Attacks

Attack Qualification

At any time after the Day 1 Visit a participant starts experiencing a non-laryngeal HAE attack, they will record the start date and time of the initial symptom of the attack in their electronic HAE diary and begin scoring the AMRA-3 assessments in the electronic device. For a non-laryngeal attack to qualify for treatment with study drug, the following criteria must be met:

- For adolescent participants only, the attack started at least 72 hours after previous study drug administration for PK assessment at the Day 1 Visit.
- The attack started at least 72 hours after treatment of a previous attack with study drug (or after the time of second dose of study drug or rescue medication, if used) or usual standard-of-care HAE medication, and participant has been symptom free for at least 24 hours.
- At least one attack symptom (skin pain, skin swelling, or abdominal pain) has reached the individual item score of ≥20 in AMRA-3.
- The participant is able to self-administer study drug while on the qualification call with the investigator (or designee) or immediately thereafter. Note that study drug must be taken within 6 hours after onset of symptoms at any location.
- The participant is able to conduct the ePRO assessments at the predefined time points right before and up to 48 hours after treatment, particularly the assessments during the first 4 hours. Participants are encouraged to enter data at all designated time points. Note that any time points missed while asleep at ≥12 hours post-dose will not be considered a protocol deviation.
- For qualification of the second attack: the Post Attack Visit after the first treated attack as well as the first qualitative interview have been completed.

Note: A non-laryngeal HAE attack not meeting all the above criteria should be treated with usual standard-of-care HAE medication (if needed) and the attack start date and time and attack end date and time should be recorded in the electronic HAE diary.

Attack Treatment

After attack qualification and while on the qualification call with the investigator (or designee), the participant will complete the pre-treatment assessments in the electronic device (including a body map of attack locations, the PGI-S, and AMRA), and then self-administer the study drug while on the qualification call with the investigator (or designee) or immediately after the qualification call. Post-treatment, the participant will record the time since most recent food intake before and after study drug administration. Study center personnel may retrain participants on use of the electronic device, at their discretion.

For participants treating qualified non-laryngeal attacks, **a second dose** of study drug is permitted ≥4 hours post first dose for the management of symptoms that are persisting or progressing after the first study drug dose.

- If at ≥4 hours after study drug administration, symptoms of a non-laryngeal attack are persisting or progressing, the participant will consult again with the investigator (or designee) via a telephone call or other live communication system to assess symptom relief and AEs, and any need for a second dose. Participants are encouraged to wait at least 4 hours after study drug administration to administer a second dose. If symptoms improve or resolve after the first study drug dose but then recur, a second dose may be considered for that same attack, provided that the recurrence is within 24 hours after initial symptom resolution, as defined by PGI-S.
- If at ≥1 hour after the second dose, symptoms persist or progress, then HAE on-demand rescue medication can be administered.
  - Note that, if needed for participant safety, participants may self-administer HAE rescue medication at any time and are to record the time of administration of HAE rescue medication in the electronic device.

If the participant does not require a second dose and/or has not contacted the investigator after 4 hours post first dose, there will be a contact between the participant and the investigator (or designee) at any time before 12 hours post first dose to confirm symptom relief and assess safety.

6.1.2.3.1.2. Laryngeal Attacks in Adolescents or Laryngeal Attacks in Adults with Breathing Difficulties or Stridor

Any laryngeal attack in adolescents or any laryngeal attack in adults that presents with breathing difficulties or stridor should be treated immediately with standard-of-care HAE medication and medical care must be sought urgently.

Laryngeal attacks, regardless of severity, in adolescent participants or severe laryngeal attacks with symptoms of breathing difficulties or stridor in adult participants carry a high risk of hypoxia and asphyxiation and should immediately be treated with standard-of-care HAE medication instead of study drug and seek medical care.

6.1.2.3.1.3. Non-Severe Laryngeal Attacks without Breathing Difficulties or Stridor (in Adults Only)

Non-severe Laryngeal Attack Qualification

At any time following the Day 1 Visit when participants start experiencing a non-severe laryngeal HAE attack **without** breathing difficulties or stridor, they will call their Investigator (or designee) immediately.

During the telephone contact, the Investigator will evaluate the presence and severity of laryngeal symptoms. Based on this evaluation, the Investigator will determine whether the attack qualifies for treatment with study drug or that it would be in the best interest of the participant to treat the attack with their usual standard-of-care HAE medication.

For a non-severe laryngeal attack to qualify for treatment with study drug, the following criteria must be met:

- For participants ≥18 years of age only, the attack started at least 72 hours after study drug administration at the Day 1 visit.
- The attack started at least 72 hours after treatment of a previous attack with study drug (or after the time of second dose of study drug or rescue medication, if used) or usual standard-of-care HAE medication, and participant had been symptom free for at least 24 hours.
    - The participant is able to conduct the ePRO assessments at the predefined time points right before and up to 48 hours after treatment, particularly the assessments during the first 4 hours. Participants are encouraged to enter data at all designated time points. Note that any time points missed while asleep at ≥12 hours post-dose will not be considered a protocol deviation.
    - For qualification of the second attack: the Post Attack Visit after the first treated attack as well as the first qualitative interview must have been completed.

If the participant cannot reach the Investigator immediately, they should treat the laryngeal attack with their usual standard-of-care HAE medication and not with study drug.

If the investigator does not consider the attack to be a laryngeal attack but a non-laryngeal attack instead, the participant will be prompted to complete the AMRA-3 assessments on their electronic device and the qualification steps as per Section 6.1.2.3.1.1 will be followed.

Laryngeal Attack Treatment

After attack qualification, the participant will complete the pre-treatment assessments in the electronic device (including AMRA-5, a body map of attack locations, and the PGI-S), and then self-administer the study drug while on the qualification call with the Investigator (or designee) or immediately after the qualification call. Post-treatment, the participant will record the time since most recent food intake before and after study drug administration.

If after study drug administration, symptoms of a laryngeal attack are persisting or progressing, the participant should follow-up with the Investigator. Note that the participant may self-administer rescue medication at any time, if needed, and record the time of administration of

rescue medication in the electronic device. A second dose of blinded study drug should not be administered for non-severe laryngeal attacks.

During the laryngeal attack treatment period, the participant must connect with the Investigator (or designee) via a telephone call or another live connection system to assess symptom relief and AEs until the resolution of the laryngeal attack; at least 1 contact must occur between 4 and 12 hours post study drug administration.

6.1.2.4. Follow up After Attacks

6.1.2.4.1. Follow up After First Treated Attack

- A qualitative interview regarding experience with HAE medications (including study drug), preferences, and non-localized symptoms typically experienced by the participant with HAE attacks (such as anxiety or fatigue), as well as health-related quality of life as measured by EQ-5D-5L, will be conducted ≥48 hours to ≤10 days following study drug administration for the first treated attack, if allowed by local regulations. The interview will be conducted by an external vendor via remote call.
- After the participant self-administers study drug for the first qualifying HAE attack, they will have either an on-site or remote Post Attack Visit ≥48 hours to ≤10 days after study drug was taken for evaluation and recording of AEs and concomitant medication use. The study drug for the second attack will be dispensed during the on-site visit or shipped from the site to the participant (if allowed by local regulation) after the attack follow-up visit has been completed and all study drug dispensed for the first attack has been returned (e.g., via courier) to the site.

  Any attacks that occur between the first attack and receipt of the study drug for the second attack must be recorded in the handheld device as a non-qualifying attack and participants should treat the attack with their usual on-demand HAE medication.

  The study drug for the second attack can only be self-administered once the Investigator (or designee) has confirmed a subsequent HAE attack is occurring that meets the qualification criteria.

6.1.2.4.2. Follow up after Second Treated Attack (End-of-Study Visit)

The End-of-Study visit includes both on-site and remote assessments.

- A qualitative interview regarding experience with HAE medications (including study drug), preferences, and non-localized symptoms associated with HAE attacks (such as anxiety or fatigue), as well as health-related quality of life as measured by EQ-5D-5L, will be conducted ≥48 hours to ≤10 days following the study drug administration for the second attack treated, if allowed by local regulations. The interview will be conducted by an external vendor via remote call. The qualitative interview following the second treated attack must take place before completion of the End-of-Study Visit.
- After the second qualifying attack is treated with study drug, an on-site End-of-Study Visit is to occur at 10 (±5) days following the attack resolution, with return of the

electronic device, return of used/unused study drug and rescue medication, and collection of safety laboratory data, vital signs, an ECG, urine pregnancy test (for females of childbearing potential), and recording of AEs and concomitant medication use.

6.1.2.5. Early Termination

For participants who discontinue from the study prematurely, final assessments (see End-of-Study Visit) will be performed 10 (±5) days after the last administration of study drug or as soon as possible after the decision to discontinue study participation is communicated to Investigator/designee. Participants who withdraw consent from further participation will be asked to attend a visit to complete final assessments as soon as possible for safety purposes. Participants who discontinue from the study prematurely will not be eligible for participation in the open label extension study.

6.2. Rationale for Study Design and Control Group

In this study, a placebo control is utilized, and randomization will be used to assign participants to a treatment sequence (active then placebo, or placebo then active). Each participant will receive both active and placebo treatment in a crossover fashion to treat 2 separate qualifying attacks; a washout of at least 72 hours must elapse between taking study drug or any other HAE medication (standard or rescue medication) for the attack to qualify for study drug administration to avoid the possibility of any carry-over effect. To best represent a scenario closer to real-world clinical practice, a heterogeneous population will be included in this study (adolescents ≥12 to <18 years of age, adults [≥18 years of age], participants not using long-term HAE prophylaxis and participants using long-term HAE prophylaxis, and multinational locations with different disease prevalence, including by gender, and standard-of-care). Therefore, a crossover design was chosen for this study to minimize inter-participant variability as each participant serves as their own control.

Retreatment of a non-laryngeal attack with deucrictibant in the absence of sufficient symptom relief (i.e., worsening, partial response, or recurrence of symptoms) at least 4 hours after the first dose is supported by clinical PK and efficacy data. If there is no symptom relief or if symptoms recur after the second dose of study drug, HAE rescue medication can be used.

Inclusion of adolescents in this Phase 3 study is justified because safety and efficacy data from the Phase 2 (PHA022121-C201) study were robust and support the initiation of the pediatric development of deucrictibant. In the study, all primary and key secondary efficacy endpoints were met, and deucrictibant was demonstrated to be well-tolerated. The majority of TEAEs reported was mild or moderate in severity, with no treatment-related SAEs, severe AEs, or AEs leading to treatment discontinuation being reported. Some adolescents experience HAE attacks at a frequency that is similar to that for adults, and they have body mass indexes approaching those of adults. Therefore, it is proper to include adolescents in this study. Additionally, inclusion of adolescents also is supported by population PK analyses conducted by the Sponsor. HAE attacks in younger children (<12 years) are rare and will be evaluated in a separate study.

The 20 mg dose of deucrictibant was chosen for this study based on results from the Phase 2 study, in which 20 mg was more effective than 10 mg, and 20 mg was equally effective to 30 mg. Population PK data support this dose for adolescent participants weighing ≥40 kg.

Global experts in HAE were consulted in the design of this study. In addition, the Sponsor undertook patient involvement in the design of the study in multiple global patient forums which informed design aspects with feedback to reduce participant burden, where possible.

6.3. Appropriateness of the Measurements

All efficacy, safety, and PK measurements in the study are widely used and generally recognized as reliable, accurate, and relevant.

6.4. Study Duration and Dates

The duration of an individual's participation will depend on the time needed for the participant to experience 2 qualifying attacks treated with study drug. Based on prior clinical study experience, the average duration is estimated to be approximately 42 weeks for each participant (but actual duration may vary from 6 to >60 weeks).

6.5. Number of Participants

Approximately 120 participants will be enrolled; see Section 12.2.

6.6. Treatment Assignment

See Section 8.4.

6.7. Dose Adjustment Criteria

No study drug dose adjustments are permitted.

6.8. Criteria for Study Termination

The study can be terminated by the Sponsor at any time for any reason. This may include reasons related to the benefit/risk assessment of participating in the study, practical reasons (including slow enrollment or business considerations), or for regulatory or medical reasons. In taking the decision to terminate, the Sponsor will always consider the participant's welfare and safety.

Should early termination be necessary, participants must be seen as soon as possible (instructions will be provided to the Investigator for contacting the participant, when the participant should stop taking drug, and when the participant should come for a final visit) and treated as a prematurely withdrawn participant. The investigator may be informed of additional procedures to be followed in order to ensure that adequate consideration is given to the protection of the participant's interests. The Investigator or Sponsor, depending on the local regulations, will be responsible for informing Institutional Review Boards (IRBs)/Independent Ethics Committees (IECs) of the early termination of the study.

The Sponsor may decide to terminate the study for the purpose of final data analysis once a sufficient number of participants have completed the treatment phase, with one attack treated with study drug in each of the 2 periods in the treatment phase (see Section 12.2). If the Sponsor

decides to terminate the study for this reason, all participants still enrolled at the time of study termination will have an End-of-Study Visit and will be eligible for continuation of treatment with deucrictibant [20 mg] in the open-label extension study.

6.9. Study Completion

The study will be considered complete when the last participant completes the End-of-Study Visit (see Section 6.8).

7. SELECTION AND WITHDRAWAL OF PARTICIPANTS

The study will enroll approximately 120 adult and adolescent participants (≥12 to ≤75 years of age with HAE. The study will target enrollment of approximately 9 adolescent participants (≥12 to <18 years of age at the time of screening with consideration of local age requirements).

The study is planned to be conducted at study sites in approximately 30 countries.

7.1. Participant Inclusion Criteria

Participants must meet all of the following criteria to be enrolled in this study:

1. Provision of written informed consent. If the participant is a minor (i.e., <18 years of age or as determined by local law), consent will be obtained from the participant's parent/legally designated representative/guardian and written assent will be obtained from the participant, per country regulations.
2. Male or female, aged ≥12 to ≤75 years at the time of providing written informed consent/assent.
3. Diagnosis of HAE based upon the following:
   a. For participants with HAE type 1/2:

   - Documented clinical history consistent with HAE (subcutaneous or mucosal, nonpruritic swelling without accompanying urticaria)
   - At least one of the following:
     - Age ≤30 years at reported onset of first angioedema symptoms
     - Family history consistent with HAE type 1/2
     - C1q performed by the central laboratory is above the lower limit of the normal range
   - Diagnostic testing results to confirm HAE type 1/2:
     - Participants on long-term prophylactic HAE therapies with C1-INH or danazol at the time of study entry must provide documentation of a historical laboratory test showing C1-INH functional level <50%.

       Note: If a confirmatory C1-INH test is required, the sample should be collected at least 5 half-lives after the last dose of C1-INH or danazol. This procedure will not be considered an interruption of stable LTP treatment.
     - For all other participants with HAE type 1/2, C1-INH functional level of <50% must be shown by chromogenic assay performed by the central laboratory as part of the screening procedures.

   b. For participants with HAE type 3:

   - Recurrent angioedema attacks with diagnostic testing results obtained during screening to confirm C1-INH function ≥50% of normal and C4 level not below lower level of normal range performed by the central laboratory

Note: The C1-INH sample should be collected at least 5 half-lives after the last dose of C1-INH therapy, if applicable.

- Must meet one of the following:
  - Documented genetic mutation associated with HAE type 3 as listed in the HAEA and WAO/EAACI Guidelines (Appendix 1)

  OR

  - If no documented mutation:
    - Clinical diagnosis with family history of HAE type 3 and an elevated BK peptide level confirmed by a commercially available assay

    AND

    - Attacks not responding to treatments with high-dose antihistamine (cetirizine 40 mg/day or equivalent high-dose second-generation antihistamine medication) and no clinical attack symptoms relief if treated with corticosteroid, montelukast, or omalizumab
- Documented effective attack symptom relief with on-demand icatibant treatment

4. History of at least 2 HAE attacks in the last 3 months before screening.
5. Experience with using standard-of-care treatment to effectively manage on-demand treatment for HAE attacks.
6. Either:
   - Is on a stable dose and regimen of long-term prophylactic therapy with one of the following medications indicated for HAE: plasma-derived C1-INH, danazol at ≤200 mg/day, anti-fibrinolytics, berotralstat, or lanadelumab, for at least 6 months before the Screening Visit and intend to remain on the same dose for the duration of the study

   OR

   - Is receiving only on-demand treatment, and has not used long-term prophylactic treatment within the following durations before the Screening Visit, as specified below:
     - Plasma-derived C1-INH, berotralstat, anti-fibrinolytics, or attenuated androgens for at least 2 weeks
     - Lanadelumab for at least 10 weeks
7. Capable of recording, without assistance, electronic HAE diary and ePRO data using an electronic device, as evidenced by the competency assessment conducted during the Screening Phase.
8. For adolescent participants aged ≥12 to <18 years: body weight ≥40 kg.
9. Female participants of childbearing potential (or who become of childbearing potential during the study) must agree to the protocol specified pregnancy testing and to be

abstinent from heterosexual intercourse or to use an acceptable contraception method, as defined in the protocol and as available locally, from enrollment until 30 days after the last study drug administration.

There are no contraceptive requirements for male participants.

10. In the opinion of the Investigator, the participant (and parent/caregiver for adolescent participants) is willing and able to comply with the protocol.

7.2. Participant Exclusion Criteria

Participants who meet any of the following criteria will be excluded from the study.

1. Any female who is pregnant, plans to become pregnant, or is breastfeeding.
2. Any diagnosis of angioedema other than HAE.
3. Any clinically significant comorbidity or systemic dysfunction (e.g., cardiovascular, gastrointestinal, renal, neurological, respiratory) that, in the opinion of the Investigator, would interfere with the participant's safety or ability to participate in the study.
4. Use of attenuated androgens for short-term prophylaxis within 2 weeks before screening.
5. Abnormal hepatic function (aspartate aminotransferase [AST] >2×upper limit of normal [ULN], alanine aminotransferase [ALT] >2×ULN, or total bilirubin >1.5×ULN, or Child-Pugh class B or C). Participants with Gilbert's syndrome, being defined as an isolated increase in total bilirubin ≤3×ULN, with AST and ALT in the normal range, will not be excluded.
6. Abnormal renal function (estimated glomerular filtration rate <60 mL/min/1.73 m^2).
7. History of alcohol or drug abuse within the previous year, or current evidence of substance dependence or abuse.
8. Has received prior on-demand HAE treatment with deucrictibant.
9. Currently participating in any other investigational drug study or receiving other investigational treatment within the last 30 days, or within 5 half-lives (whichever is longer) of the time of randomization.
10. Prior gene therapy for any indication at any time.
11. Use of concomitant medications with systemic absorption that are strong inhibitors of CYP3A4 (such as clarithromycin, itraconazole, ketoconazole, and ritonavir) or strong inducers of CYP3A4 (such as carbamazepine and phenytoin) within the last 30 days or within 5 half-lives (whichever is longer) of the time of randomization.
12. Known hypersensitivity to deucrictibant or any of the excipients of study drug.

7.3. Participant Withdrawal Criteria

Participants are free to withdraw from participation in the study at any time upon their request. The investigator and/or sponsor may withdraw a participant from the study for any of the following reasons:

- Pregnancy
- A protocol or eligibility violation occurs
- A serious or intolerable adverse event occurs
- A clinically significant change in a laboratory parameter occurs
- Other medical condition or situation occurs that negatively impacts that benefit-risk such that continued participation in the study would not be in the best interest of the participant
- Sponsor termination of the study
- Investigator considers it is in the participant's best interest to withdraw
- Participant request

A participant will be considered lost to follow-up if he or she fails to return for a scheduled visit and is unable to be contacted by the study site staff.

The reason for participant discontinuation or withdrawal from the study will be recorded in the eCRF and documented in the participant record.

Participants who are discontinued from study drug prematurely will be requested to complete an End-of-Study Visit at 10 (±5) days after the last treatment administration or decision to discontinue study, whichever is later.

7.3.1. Replacement Policy

Participants will not be replaced in this study.

8. TREATMENT OF PARTICIPANTS

8.1. Description of Study Drug

A description of deucrictibant soft capsules is presented in Table 5.

Table 5: Investigational Product

Product Name:	Deucrictibant soft capsule
Dosage Form:	White, oval, soft capsule
Unit Dose	Each soft capsule contains 20 mg of deucrictibant
Route of Administration	Oral
Physical Description	White opaque, oval soft capsule

8.2. Concomitant Medications

Prior non-HAE therapies used by the participant within 30 days before the screening visit must be documented as such in the eCRF.

All concomitant HAE and non-HAE treatments (including but not limited to all prescription, over-the-counter medications, vitamins and supplements, herbal treatments, non-drug supportive therapies and procedures) administered to participant from the time of signing the ICF through the final study contact (i.e., End-of-Study visit) are considered concomitant therapies and must be documented as such in the eCRF, including the start and stop dates and times, dose, frequency, and indication.

8.2.1. Rescue Medication

Where feasible, commercially available icatibant will be provided as a rescue medication to be used if needed if the study drug does not provide sufficient symptom relief.

Icatibant will be provided as pre-filled syringes of 3 ml, containing 30 mg solution for sc injection. Each mL of the clear and colorless liquid solution contains 10 mg of icatibant. Additional information on physical and chemical properties of icatibant may be found in the SmPC or national label. The SmPC or national label also provides details for storage and administration.

The participant can alternatively use another type of approved, prescribed, readily available HAE rescue medication.

Participants may self-administer rescue medication at any time during the study drug-treated HAE attack if needed for symptom relief, and will inform the investigator, as well as record the time of administration of rescue medication in the electronic HAE diary.

(For participants with non-laryngeal attacks, it is preferred that a second study drug dose be administered before any rescue medication. However, if needed, participants may self-administer

HAE rescue medication at any time and record the time of administration of HAE rescue medication in the electronic device.)

8.2.2. Allowed HAE Medication

All HAE medications (on-demand and/or prophylactic therapy) taken by the participant within 6 months before the screening visit will be documented in the eCRF with start date, frequency, stop date or ongoing, and reason for discontinuation if stopped.

During the study, usual standard-of-care HAE medication (products approved to treat acute HAE attacks such as icatibant, C1-INH, ecallantide) should be administered immediately for any severe laryngeal attack that presents with breathing difficulties or stridor and be documented in the eCRF.

Short-term (one-time use) prophylactic therapy for HAE attacks following local standard before medically indicated procedures (e.g., surgery, dental procedure) is allowed, but the medication needs to be washed out (i.e., at least 72 hours or >5 × half-life, whichever is longer) before study drug administration.

Refer to Section 7.1 for details regarding the use of long-term prophylactic therapy during the study.

8.2.3. Contraception

Following completion of the EFD studies in pregnant animals, deucrictibant is considered a study drug with unlikely human teratogenicity/fetotoxicity in early pregnancy according to the Clinical Trials Coordination Group recommendations (CTCG, 2024).

Female participants of childbearing potential require the use of medically accepted, highly effective contraceptive measures from enrollment until 30 days after the last study drug administration.

The following birth control methods are considered highly effective:

- Progestogen-only hormonal contraception associated with inhibition of ovulation: oral, injectable, or implantable
- Intrauterine device (IUD, all types)
- Intrauterine hormone-releasing system (IUS)
- Bilateral tubal occlusion
- Vasectomized partner (provided that partner is the sole sexual partner and that the vasectomized partner has received medical assessment of the surgical success.)
- Sexual abstinence, where it is the anticipated usual lifestyle of the participant

Females of non-childbearing potential, defined as prepubertal, surgically sterile (status post hysterectomy, bilateral salpingectomy, or bilateral oophorectomy), or post-menopausal (defined as no menses for at least 12 months without an alternative medical cause and an FSH test result indicative of post-menopausal status) do not require contraception during the study.

There are no contraceptive requirements for male participants.

Note: Not all methods acceptable for use during participation in this study are approved by local regulations. The selection of a contraceptive method should be based on the approved methods in the participant's country.

8.2.4. Prohibited Concomitant Medications

8.2.4.1. Prohibited Therapy

The following therapies are prohibited during study participation:

- Systemic corticosteroids, antihistamines, narcotics, and other over-the-counter pain relief medications are prohibited for on-demand treatment of an HAE attack during the HAE attack period but can be used to treat intercurrent illnesses between HAE attacks if needed. Regular use of systemic steroids or antihistamines at a stable dose for other indications (e.g., asthma, allergic rhinitis) and use of topical, inhaled, and intranasal corticosteroids and antihistamines is permitted.
- Concomitant medications with systemic absorption that are strong inhibitors of CYP3A4 (such as clarithromycin, itraconazole, ketoconazole, and ritonavir) or strong inducers of CYP3A4 (such as carbamazepine and phenytoin).
- Angiotensin converting enzyme inhibitors
- Estrogen-containing medications with systemic absorption
- Any other investigational drug or device

8.2.5. Restrictions

8.2.5.1. Fluid and Food Intake

No restrictions for fluid and food intake apply during the study. However, the time of the most recent food intake before and after each study drug dose will be documented.

8.2.5.2. Participant Activity Restrictions

No restrictions on participant activity apply during the study.

8.3. Treatment Compliance

Day 1 Visit

Adolescents only (≥12 to <18 years): deucrictibant for the Day 1 visit will be administered for PK assessment at the study site under the supervision of the study staff, thus ensuring treatment compliance.

Adult participants: No PK assessment with deucrictibant for participants ≥18 years of age at the Day 1 Visit.

Treatment Phase

Adults and adolescents: Participants will record the dates and times of administration of the study drug in the electronic device.

8.4. Randomization and Blinding

Adolescents only (≥12 to <18 years): Each adolescent participant will be provided a single, oral dose of deucrictibant (20 mg) for administration for PK assessment during the Day 1 Visit.

Adults and adolescents: Randomization to a crossover treatment sequence of blinded study drug (active-placebo or placebo-active) will be done via an interactive response technology (IRT) system. Randomization will be stratified by age group (≥12 to <18 years, ≥18 years) and by use of HAE medication for long-term prophylaxis (Yes/No).

The study population will include approximately 30% of participants who are using long-term prophylactic medication for HAE globally; in the United States (including Puerto Rico), enrollment is targeted to be approximately 50% of participants using long-term prophylactic medication.

This target proportion of participants using long-term prophylactic medication is intended to increase an opportunity to assess on-demand treatment with deucrictibant in participants with severe attacks. Severe attacks, including laryngeal attacks, appear to be observed more frequently in patients who are not receiving long-term prophylactic therapy compared to those receiving such therapy, as suggested by global HAE experts consulted in the design of this study, and observed in the Phase 2 study with deucrictibant for prophylactic treatment (Study PHA022121-C301). In that study, during deucrictibant prophylactic treatment, all breakthrough attacks were mild or moderate and there were no severe attacks reported. In addition, the global proportion of 30% for participants with long-term prophylactic therapy for enrollment is aligned with the enrollment status from a recently completed global Phase 3 oral on-demand treatment study, in which approximately 22% of participants enrolled were receiving long-term prophylactic therapy (Riedl, 2024). The higher proportion of approximately 50% in the United States reflects the fact that there are a relatively greater number of patients with HAE receiving long-term prophylactic therapy in the United States than in the rest of world due to variable availability of such medications.

The packaging and labeling are blinded.

Access to treatment sequence assignment will be available if the Investigator deems it necessary to break the study blind in the interest of a participant's medical safety. The Sponsor requests that the Investigator contacts the Medical Monitor to discuss the situation which has arisen and resulted in the need for unblinding of the treatment assignment. However, unblinding procedures can be initiated by the investigator without the need to discuss unblinding in advance with the Medical Monitor in an emergency, potentially life-threatening situation.

Unblinding can also be performed to meet regulatory reporting obligations, or if warranted during scheduled safety reviews. The unblinding procedures are detailed in the safety management plan.

9. STUDY DRUG MATERIALS AND MANAGEMENT

9.1. Study Drug

Study drug refers to both deucrictibant and the matching placebo soft capsule, administered as blinded study drug.

9.1.1. Active Study Drug

Active study drug is a soft capsule for oral administration containing 20 mg deucrictibant.

9.1.2. Placebo

To maintain the study blind during the attack Treatment Phase, matching placebo for oral administration will be used.

9.1.3. Rescue Medication

Where feasible, commercially available icatibant will be provided for use as rescue medication if the study drug does not provide sufficient symptom relief. Commercially available icatibant will be labeled according to the applicable regulations for the country in which it is used. The participant can alternately use another readily available, prescribed HAE medication as rescue medication that is approved in their country for the on-demand treatment of HAE.

9.2. Study Drug Packaging and Labeling

The study drug (deucrictibant or placebo) is packaged according to current Good Manufacturing Practice.

The study drug is packaged in aluminium/aluminium blisters. Each blister will be sealed into a wallet. The wallet will be labeled with a booklet label and rendered tamper evident.

Each wallet will contain one single dose, which consists of 1 soft capsule (deucrictibant or placebo).

Each participant will receive 2 study drug wallets for each attack. The participant will have to return any study drug after the first attack to the investigational site before dispensation of study drug for the second attack can occur.

9.3. Study Drug Storage

The investigator or designee has overall responsibility for ensuring that the study drug is stored in a secure, limited-access location. Access to the study drug should be restricted to designated study personnel.

Participants will be instructed on appropriate storage of the study drug at their home location.

The study drug should not be stored above 25ºC (do not freeze). Further details on study drug storage will be provided in the pharmacy instructions provided separately.

9.4. Administration

Day 1 Visit

Adolescents only (≥12 to <18 years) will administer a single deucrictibant soft capsule (20 mg) on Day 1 for the purpose of assessing PK and safety of deucrictibant.

Treatment Phase

Adults and adolescents: Participants will administer study drug (deucrictibant or placebo) according to their randomized crossover treatment sequence for on-demand treatment of 2 qualifying HAE attacks. Participants with qualifying non-laryngeal attacks may administer a second study drug dose in case symptoms are persisting or progressing, as described in Section 6.1.2.3.1.1.

9.4.1. Selection and Timing of Dose for Each Participant

Day 1 Visit

Adolescents only (≥12 to <18 years) will administer a single deucrictibant soft capsule (20 mg) in a non-attack state on Day 1 at the study site. The study drug will be taken with approximately 200 mL of water.

Treatment Phase

Adults and adolescents: Blinded study drug will be dispensed to each participant according to their randomized crossover treatment sequence for use during the Treatment Phase. Participants will self-administer oral doses of blinded study drug in a crossover fashion: a 20 mg deucrictibant for the first qualifying attack followed by placebo for the second qualifying attack or the reverse order. Participants with qualifying non-laryngeal attacks may self-administer a second study drug dose in case symptoms are persisting or progressing, as described in Section 6.1.2.3.1.1.

The study drug should be taken with 200 mL of water and can be taken with or without meals. The timing of the most recent food intake before and after each study drug dose will be documented in the electronic HAE diary.

9.5. Study Drug Accountability

The study drug must be dispensed under the supervision of the Investigator or designee using the IRT system. The Investigator or designee pharmacist will maintain up-to-date accountability records recording receipt, dispensing, return, and destruction (Section 9.6) of all clinical study supplies. Local regulations must be adhered to.

Any damaged or unusable study drug in a given shipment will be documented in the study files and recorded in IRT. The Investigator must notify the Sponsor of any damaged or unusable study drug that was supplied to the Investigator's site.

Any medication errors, including incorrect dosage, use of expired study drug, storage temperature excursions, administration errors, etc, should be recorded as protocol deviations and the Sponsor should be immediately informed.

9.6. Study Drug Handling and Disposal

All unused study drug and empty/used study drug packaging will either be returned to the Sponsor (or nominated facility) or can be destroyed on site (according to local regulations) or by an authorized destruction unit following written agreement with the Sponsor. All correspondence related to study drug destruction must be filed with the study accountability records.

10. ASSESSMENT OF EFFICACY

The ePRO assessments include the PGI-C, PGI-S, and AMRA and are to be completed at the timepoints indicated in the Schedule of Events (Table 4). The PROs must be completed pre-treatment (with the exception of PGI-C) at the time the investigator is consulted to confirm qualification of the attack for treatment with study drug, and at 15 and 30 minutes after self-administration of study drug, then every 30 minutes through 4 hours post-dose, and at 5, 6, 7, 8, 9, 10, 11, 12, 14, 16, 18, 20, 22, 24, 36, and 48 (+6) hours post-dose. The time window between assessment time points will be the midpoint between adjacent scheduled time points. If a second dose and/or rescue medication are needed, PROs should be additionally recorded just before taking the second dose or rescue medication. The time of dosing with the second dose of study drug or rescue medication will be recorded and the participant will continue recording ePRO assessments at each of the remaining time points.

Participants are encouraged to enter data at all designated time points. Note that any time points missed while asleep at ≥12 hours post-dose will not be considered a protocol deviation.

Provisions will be made for back-up method(s) in case of device malfunction; details will be in the training material provided to the participant.

10.1. Patient Global Impression of Change

The PGI-C is designed to evaluate the change in the attack symptoms from pre-treatment (the last symptom assessment before study drug administration) over time with a 7-point response scale. Participants will be asked to choose the response that best describes currently at the time of assessment the overall change in their HAE attack symptoms since they took the study drug:

- Much better
- Better
- A little better
- Same
- A little worse
- Worse
- Much worse

The PGI-C will be completed post-treatment at the same time points as AMRA. If a second dose of study drug and/or rescue medication is needed, the PGI-C should be recorded just before taking the second dose of study drug or rescue medication.

10.2. Patient Global Impression of Severity

The PGI-S is designed to evaluate the severity of attack symptoms on a 5-point response scale. Participants will be asked to choose the response below that best describes the severity of their HAE attack symptoms currently at the time of assessment:

- None
- Mild
- Moderate
- Severe
- Very Severe

The PGI-S will be completed at the same time points as AMRA. If a second dose of study drug and/or rescue medication is needed, the PGI-S should be recorded just before taking the second dose of study drug or rescue medication.

10.3. Angioedema Symptom Rating Scale

The AMRA-3 scale allows scoring of the 3 major HAE symptoms (skin swelling, skin pain, and abdominal pain) on a scale between 0 (No swelling/No pain) and 100 (Worst possible swelling/Worst possible pain). The AMRA-3 is a composite score of the 3 major HAE symptoms. If the participant has symptoms of a non-laryngeal attack, AMRA-3 will be completed thereafter.

For non-severe laryngeal attacks, the AMRA-5 allows scoring of the 5 major HAE symptoms (skin swelling, skin pain, abdominal pain, voice change, and difficulty swallowing) on a scale between 0 (No skin swelling, skin pain, abdominal pain, voice change, or difficulty swallowing) and 100 (Worst skin swelling, skin pain, abdominal pain, voice change, or difficulty swallowing). If the participant has symptoms of a non-severe laryngeal attack, AMRA-5 will be completed thereafter.

10.4. Qualitative Interviews (Exploratory)

Qualitative interviews regarding experience with HAE medications (including double-blinded study drug), treatment preferences, and non-localized symptoms the participant typically experiences with HAE attacks (such as anxiety or fatigue), impairment of daily activities, as well as health-related quality-of-life as measured by EQ-5D-5L, will be conducted ≥48 hours to ≤10 days following each of the 2 attacks treated with study drug, if allowed by local regulations. Interviews will be conducted by an external vendor via remote call. The qualitative interview following the second treated attack must take place ≥48 hours to ≤10 days following the study drug administration and before completion of the End-of-Study visit.

10.5. AMRA Interview

An interview to generate evidence on the content validity of the AMRA-3/AMRA-5 will be collected in a subset of participants at selected sites within 72 hours after the Day 1 Visit. The interview will be conducted by an external vendor via remote call.

11. ASSESSMENT OF SAFETY AND PHARMACOKINETICS

11.1. Informed Consent

At the Screening Visit, a signed ICF, and assent form, if applicable, must be obtained from each participant or participant's parent/legally designated representative/guardian before performing any study-related procedures. Each participant will be given both verbal and written information describing the nature and duration of the clinical study. Further details on participant consent are provided in Section 15.2.

If an adolescent participant reaches the age of 18 years during study participation, the participant will be required to provide written informed consent in order to continue study participation, in accordance with local regulations.

11.2. Demographics and HAE Characteristics

At screening, demographic information, including year of birth, gender, and, where permitted, race and ethnicity, will be captured in the eCRF for each participant. Demographic data may be used to facilitate subgroup analyses of study data.

In addition, HAE characteristics, including the year of diagnosis, family history, the type of HAE, the number of attacks in the last 3 consecutive months, and, for those completing the qualitative interview, the type of non-localized symptoms typically associated with their HAE attacks will be collected.

To confirm the HAE diagnosis, assessment of C1-INH functional level and a C1q level above the lower limit of the normal range must be documented, per the Inclusion Criterion 3 (Section 7.1).

11.3. Medical History

At screening, existing relevant medical history (current or resolved at the time of signing the ICF) will be recorded for the following body systems: head/eyes/ears/nose/throat, respiratory, cardiovascular, gastrointestinal, genitourinary, musculoskeletal, neurological, endocrine/metabolic, blood/lymphatic, dermatologic, psychiatric, allergy/immunology, and other.

11.4. Physical Examination

A physical examination will be conducted at screening. The following body systems will be examined: gastrointestinal system; head/eyes/ears/nose/throat; endocrine (including thyroid); respiratory system; cardiovascular system; integumentary system (skin); and nervous system (central and peripheral). Clinically significant findings should also be documented on the Medical History eCRF. Height (at screening) and body weight (screening and End-of-Study visit) will also be recorded.

11.5. Vital Signs

Vital signs include blood pressure, heart rate, and body temperature. Vital signs will be measured at screening and at other time points specified in the schedule of events. Blood pressure (systolic and diastolic) and heart rate will be taken after the participant has rested for at least 5 minutes.

Blood pressure measurements must be obtained with an appropriate cuff size and with the participant's arm supported at the level of the heart. It is acceptable to obtain a heart rate from the blood pressure instrument or the ECG machine.

11.6. Electrocardiograms

A digital 12-lead ECG (recorded in triplicate) will be performed and recorded, after the participant has rested in the supine position for at least 5 minutes at the Screening Visit, and at the End-of-Study Visit. Central reading of all recorded ECGs will be performed by the ECG vendor. ECGs will be evaluated by the Investigator based on the central reading. If the reading is abnormal, the Investigator will assess whether it is "Clinically significant" or "Not clinically significant", and recorded in the eCRF.

11.7. Clinical Laboratory Tests

11.7.1. Safety Laboratory Parameters

Safety laboratory testing will be performed at screening and at other time points specified in the Schedule of Events (Table 4). Clinical laboratory tests for hematology, biochemistry, coagulation, and urinalysis are indicated in Table 6. Females of childbearing potential must have urine pregnancy testing performed at the Screening Visit and at other time points specified in the Schedule of Events. Serum pregnancy and/or more frequent pregnancy testing will be performed if required by local regulations or in case of suspected pregnancy.

Unscheduled clinical laboratory testing is permitted if deemed necessary for safety by the Investigator and/or Sponsor.

Additional safety sampling may be performed and processed by a local laboratory at the same time points as the central laboratory, as per the Schedule of Events. Based on Investigator determination or local requirement for participant safety, additional samples may be assessed during the study by a local laboratory, in parallel with the central laboratory.

Participants will be in a seated or supine position during blood collection. Blood and urine samples will be processed by a central laboratory, with the exception of urine pregnancy testing. The Investigator will classify all laboratory values outside the normal range of the central laboratory as "Clinically significant" or "Not clinically significant".

Table 6: List of Safety Laboratory Tests

Hematology:

- Hematocrit
- Hemoglobin
- Mean corpuscular hemoglobin
- Mean corpuscular hemoglobin concentration
- Mean corpuscular volume
- Platelet count
- Red blood cell count
- White blood cell count with differential count (absolute and %)

Urinalysis:

- Macroscopic:
- Bilirubin
- Color
- Glucose
- Ketones
- Leukocyte esterase
- Nitrite
- Occult blood
- pH
- Protein
- Specific gravity
- Urobilinogen
- Microscopic examination (only if macroscopic is abnormal)

Urine pregnancy test (hCG) (only for females of childbearing potential)

Serum Chemistry:

- Albumin
- Alkaline phosphatase
- ALT
- AST
- Blood urea nitrogen
- Bicarbonate
- Calcium
- Chloride
- Creatinine
- Creatine kinase
- Gamma-glutamyl transferase
- Glucose
- Lactate dehydrogenase
- Magnesium
- Phosphorus
- Potassium
- Sodium
- Total bilirubin
- Direct bilirubin
- Total cholesterol
- Total protein
- Uric acid

Coagulation:

- Prothrombin time
- INR

Abbreviations: AST = aspartate aminotransferase; ALT = alanine aminotransferase; hCG = human chorionic gonadotropin; INR = international normalized ratio.

Refer to the Laboratory Manual for details regarding sample storage and retention.

11.8. Pharmacokinetic Sample Collection, Storage, and Shipping

Plasma samples for PK evaluation will be collected from adolescent participants only (≥12 to <18 years of age). On the Day 1 Visit, samples will be collected pre-dose, and post-dose at 15 minutes (±5), 30 minutes (±5), 1 hour (±10 minutes), 2 hours (±10 minutes), 4 hours (±15 minutes), and 8 hours (±15 minutes).

Collected plasma samples are to be stored as specified in the laboratory manual. Plasma samples will be transferred frozen to the bioanalytical laboratory contracted by the Sponsor.

Plasma will be analyzed to determine concentrations of deucrictibant and metabolites using validated liquid chromatography-mass spectrometry/mass spectrometry methods under the supervision of the sponsor.

Refer to the Laboratory Manual for details regarding sample storage and retention.

11.9. Dispensing and Accountability of Study Drug

See Section 9.5.

11.10. Electronic HAE Diary

The participant will access the HAE diary on a daily basis via the electronic device and respond to prompts about the occurrence of an HAE attack, study drug intake, rescue medication intake, whether a meal was consumed prior and after study drug intake, and other relevant information. Meal intake information will be recorded for the first study drug dose for the attack and again if the participant requires a second dose of study drug. When the participant records in the HAE diary the start time of an HAE attack, the diary will query whether the participant perceives laryngeal symptoms with breathing difficulties or stridor and, if yes, will provide a reminder to the participant to immediately take their usual standard-of-care HAE medication and seek medical care.

11.11. Adverse and Serious Adverse Events

11.11.1. Definition of Adverse Events

11.11.1.1. Adverse Event

An adverse event (AE) is defined as any untoward medical occurrence in a patient or clinical investigation participant administered a pharmaceutical product and which does not necessarily have to have a causal relationship with this treatment. An AE can therefore be any unfavorable and unintended sign, symptom, or disease temporally associated with the use of a medicinal product, whether or not considered related to the medicinal product.

Baseline medical history/conditions, abnormal physical examination findings or diagnostic/laboratory values present at the screening visit, are not considered to be AEs, and should be recorded in the eCRF as baseline medical history. Events with a start date after the signing of the ICF are considered AEs, and any worsening in severity, or an increase in frequency, or change of concomitant medication for management of a pre-existing baseline medical condition is considered to be an AE. For the purpose of this protocol, HAE attacks and their associated symptoms will not be defined as AEs, unless they meet the criteria for an SAE. Prescheduled hospitalizations for non-emergency procedures will not be considered SAEs.

11.11.1.2. Serious Adverse Event

An SAE or reaction is any untoward medical occurrence that at any dose:

- Results in death
- Is life-threatening (at the time of the event)

NOTE: The term “life-threatening” in the definition of “serious” refers to an event in which the participant was at risk of death at the time of the event; it does not refer to an event which hypothetically might have caused death if it were more severe.

- Requires inpatient hospitalization or prolongation of an existing hospitalization (note: a pre-existing condition that requires planned hospitalization is not considered as SAE)
  - Hospitalization for elective treatment of a pre-existing condition (i.e., existed before signing ICF) that did not worsen from baseline is not considered a criterion of seriousness, per se. Elective hospitalizations will therefore be reviewed for seriousness by the sponsor on a case-by-case basis, and with investigator input.
- Results in persistent or significant disability/incapacity
- Is a congenital anomaly/birth defect
- Is an important medical event that may jeopardize the participant or may require intervention to prevent one of the other outcomes listed above

Important medical events that may not result in death, be life threatening, or require hospitalization may be considered an SAE when, based upon appropriate medical judgment, they may jeopardize the participant or may require medical or surgical intervention to prevent one of the outcomes listed in the definition.

11.11.1.3. Clinically Significant Abnormalities

A clinically significant abnormality in a clinical test result, including laboratory test results, vital signs, and ECG parameters, should be reported as an AE unless it is a component of a reached diagnosis, in which case the reached diagnosis is the AE. Clinically significant abnormalities are defined as abnormalities that require investigation and/or treatment. Repeating an abnormal test is not considered sufficient for clinical significance.

11.11.2. Relationship to Study Drug

All AEs/SAEs (also referred to as “events”) will be assessed to determine their relationship to the study drug (causality). The investigator must assess the causality using the binary choices of “Related” or “Not Related," and by considering the rationales listed below. Events should be considered “Related” only if there is a reasonable possibility of a causal relationship to the study drug. These rationales allow the investigator to more clearly support their own assessments. All AEs and SAEs should be reported, regardless of causality, including events that are thought to be "Related" to targeted disease and/or population.

Examples of rationales for a **Not Related** assessment may include:

- Event attributed to concomitant medication or disease
- Event not reasonably temporally associated with study drug administration
- Negative de challenge and/or negative rechallenge

Examples of rationales for a **Related** assessment may include:

- Temporal relationship of event to investigational medicinal product (IMP) exposure
- Event is known to be associated with the IMP or drug class
- Event improved on discontinuation or dose reduction of IMP
- Event reoccurred on rechallenge with study drug
- Event cannot be explained by disease or other drugs
- Biological plausibility

11.11.3. Recording Adverse Events

The Investigator or designee will be responsible for assessing AEs and SAEs. Reporting of AEs and SAEs (events) may result from spontaneous reports by the participant, general questioning by the Investigator or designee, clinically significant abnormalities found during physical examination, or clinically significant findings from vital signs, ECGs, or diagnostic test results including laboratory tests. When assessing potential AEs, questioning should be open-ended, without leading the participant.

The Investigator is responsible for monitoring the safety of participants and for providing appropriate medical care. In addition, the Investigator is responsible for alerting the sponsor or its designee to any AEs, SAEs, or findings that seem unusual (e.g., an abnormal laboratory test result), regardless of its relationship to study drug, and even if it may be considered to be an unanticipated beneficial effect.

AEs and SAEs should be described in acceptable medical terminology using a single reached diagnosis (versus signs and symptoms of the reached diagnosis). Abnormal diagnostic test results that support and are a component of the reached diagnosis should not be recorded (use the reached diagnosis). However, if a diagnosis is initially unknown or uncertain, any signs and symptoms (or diagnostic test results) must be recorded. As soon as diagnosis is known, the event term(s) must be adjusted to the final reached diagnosis.

Other event information to be collected includes (but is not limited to): onset date of the event, assessment of Common Terminology Criteria for Adverse Events (CTCAE) grade/severity, relationship to medicinal product (i.e., causality assessment), action taken with study drug, outcome of the event, and date of resolution/stabilization of the event.

See Section 11.11.4 for information regarding follow-up for AEs and SAEs.

The Sponsor/designee retains the right to request additional information later for any participant with ongoing AEs/SAEs that are present at the End-of-Study Visit, if judged necessary.

11.11.3.1. Timing

All AEs and SAEs are to be recorded in the eCRFs from the time of signing informed consent/assent until the End-of-Study Visit. In certain situations, AE and SAE information is collected after the End-of-Study Visit (as outlined in Section 11.11.3).

Events occurring after the participant's first dose of study drug will be considered TEAEs.

11.11.3.2. Severity

The National Cancer Institute (NCI) CTCAE version 5.0 (NCI 2017) is a descriptive terminology which can be utilized for event/AE reporting. Adverse events (also referred to as "events") are classified by CTCAE grade/severity.

The CTCAE incorporates certain elements of the Medical Dictionary for Regulatory Activities (MedDRA) terminology. The System Organ Class (SOC), is the highest level of the MedDRA hierarchy, is identified by anatomical or physiological system, etiology, or purpose (e.g., the SOC called "Investigations" is used for laboratory test results). An AE is a term that is a unique representation of a specific event used for medical documentation and scientific analyses. Each CTCAE term is a MedDRA lowest level term.

Per NCI CTCAE, grade refers to the severity of the event. The CTCAE grade/severity should be evaluated and recorded on the eCRF according to the NCI CTCAE, version 5.0 (NCI 2017).

If an AE description is not graded according to the CTCAE grading system, the Investigator will assess the severity of AEs based on the following CTCAE grading definitions:

- Grade 1: Mild; asymptomatic or mild symptoms; clinical or diagnostic observations only; intervention not indicated
- Grade 2: Moderate; minimal, local or noninvasive intervention indicated; limiting age- appropriate instrumental activities of daily living (ADL)*.
- Grade 3: Severe or medically significant but not immediately life-threatening; hospitalization or prolongation of hospitalization indicated; disabling; limiting self-care ADL**.
- Grade 4: Life-threatening consequences; urgent intervention indicated
- Grade 5: Death related to AE
  - Not all Grades are appropriate for all AEs. Therefore, some AEs are listed with fewer than five options for Grade selection.

* Instrumental ADL refers to preparing meals, shopping for groceries or clothes, using the telephone, managing money, etc.

** Self-care ADL refers to bathing, dressing, and undressing, feeding self, using the toilet, taking medications, and not bedridden.

11.11.4. Adverse Event Follow-up

AEs should be followed until the event resolves or until the Investigator assesses it as chronic or stable (i.e., resolved with sequelae), the participant withdraws from the study, the participant is lost to follow-up, or the participant dies. For AEs that are ongoing or resolving at the time of the End-of-Study Visit, attempts will be made by the Investigator to collect follow-up information, in particular for all SAEs and all Grade ≥3 AEs.

AE data will be recorded in the eCRF up to the time of database lock, prior to which the status of all outstanding ongoing AEs will be reviewed and assessed by Sponsor or designee. As considered appropriate by the Sponsor or designee, follow-up information will continue to be

required from the Investigator on specific AEs (and in any case all SAEs considered related to the study drug), although related follow-up details may not be recorded in the CRF after database lock.

As per regulations, if the Investigator becomes aware of an SAE after the End-of-Study Visit and the SAE is considered by the Investigator to be at least possibly related to the study drug, the SAE will be reported to the Sponsor in accordance with Section 11.11.5.

11.11.5. Reporting Adverse Events

All SAEs after signing of the informed consent through the End-of-Study Visit must be reported to the Sponsor/designee by the Investigator within 24 hours of the Investigator's knowledge of their occurrence regardless of causality.

The SAE must be reported by immediately completing the appropriate fields of the eCRFs using acceptable medical terminology, written in English, and must include a causality assessment (relationship to study drug). Simultaneously, any other applicable eCRF pages, including relevant AE and concomitant medications eCRF pages, must also be completed. Supporting source documents such as hospital discharge summaries may be provided via the SAE reporting contact information provided below.

If the electronic data capture (EDC) system is unable to be accessed, the SAE Reporting contact information listed below should be utilized. A paper SAE Report Form should be completed and sent within 24 hours. The paper SAE report form should be sent by e-mail or fax to:

Medpace Clinical Safety

SAE Reporting Telephone Line:	+1-800-730-5779, dial 3 or +1-513-579-9911, dial 3 (the Americas) +49 89 89 55 718 44 (EU) +81-3-4563-7000, dial 3 (Asia Pacific)
Safety Fax:	+1-866-336-5320 or +1-513-570-5196 (the Americas) +49 89 89 55 718 104 (EU)
Email:	medpace-safetynotification@medpace.com

If a paper SAE Report Form was utilized, the original paper form, together with the fax confirmation sheet (if applicable), must be retained at the study site. If the initial SAE report is made verbally, via email, or by telephone, a written form of the SAE must follow within 24 hours (via eCRF entry or paper SAE Report Form). The Investigator will be requested to provide detailed information regarding the event at the time of the initial report.

Each SAE should be recorded as a single reached diagnosis on the SAE report form and AE eCRF. Accompanying signs or symptoms (e.g., physical exam findings, abnormal diagnostics test results, ECG findings) should not be reported as additional SAEs. However, if initially the diagnosis is unknown, the signs and symptoms should be reported. As soon as a reached

diagnosis causing the signs and symptoms is known, the event term(s) should be amended to the final reached diagnosis. If there are multiple signs and symptoms, all related signs and symptoms should be collapsed into the final reached diagnosis and this will require inactivation of previously reported AEs.

For all SAEs occurring during the study, the Investigator must submit follow-up reports to the Sponsor or its designee regarding the participant's subsequent course until the SAE has resolved, or until the condition is chronic or stable (resolved with sequelae), the participant withdraws from the study, the participant is lost to follow-up, or the participant dies. All follow-up information should be reported to the Sponsor/designee within 24 hours of awareness.

The Pharvaris Medical Lead will ultimately be responsible for determining whether an SAE is expected or unexpected utilizing the current reference safety information that is in place at the time of the event's occurrence and assessment. An SAE will be considered unexpected if the nature, grade/severity, or frequency of the event is not consistent with the list of expected adverse reactions.

11.11.6. Reporting Suspected Unexpected Serious Adverse Reactions

Following notification from the Investigator, the Sponsor and/or designee (i.e., CRO) will report AEs which are both serious and unexpected (i.e., not previously described in the reference safety information in the approved IB) and that are assessed as related to the study drug to the FDA, European Medicines Agency (EMA)/EudraVigilance, other concerned Regulatory Authorities, IRB/IECs and Investigators within the required timelines as specified in 21 Code of Federal Regulations (CFR) Part 312.32, Regulation (EU) No. 536/2014, the European Commission's "Detailed guidance on the collection, verification, and presentation of adverse event/reaction reports arising from clinical trials on medicinal products for human use" (CT-3, June 2011), ICH GCP Guidelines, and other applicable national regulatory requirements. Typically, fatal and life-threatening adverse reactions must be reported immediately and not later than 7 calendar days and all other unexpected serious adverse reactions must be notified immediately and no later than 15 calendar days. All other SAEs whether related to the study drug will be reported to the concerned Regulatory Authorities, and IRB/IECs at least annually in a summary format (e.g., Development Safety Update Report).

Additionally, events may occur during the clinical development of a study drug which do not fall within the definition of a suspected unexpected serious adverse reaction (SUSAR) and thus are not subject to the reporting requirements for SUSARs, even though they may be relevant in terms of participant safety. Examples are new events (or findings) related to the conduct of a study or the development of a study likely to affect the safety of participants such as:

- A significant hazard to the participant population such as lack of efficacy of a study drug used for the treatment of a life-threatening disease
- A major safety finding from a newly completed animal study (such as carcinogenicity)
- A temporary halt of a study for safety reasons if the study is conducted with the same study drug in another country by the same Sponsor, where relevant for the safety of participants

These events (or findings) are not to be reported as SUSARs, but they might require other action, such as urgent safety measures and their notification, substantial amendments, or early termination of the study, and will be reported in accordance with applicable local regulations and guidelines.

11.11.7. Reporting of Pregnancy

Any female participant who has taken study drug and who becomes pregnant during the study will interrupt administration of study drug, with no further doses of study drug taken nor daily diary completed until resolution of the pregnancy. The pregnancy will be followed until the pregnancy's outcome to assess for any potential occurrence of congenital anomalies or birth defects (discovered in utero or at birth). The participant may discontinue the study or study drug may be resumed under the auspices of the current study after resolution of the pregnancy, or if the current study is closed, under the auspices of the long-term extension study, provided all entrance criteria are met for that study.

While pregnancy is not considered an AE per se, pregnancies must be reported immediately to the Sponsor or designee using the Pregnancy Report Form and following SAE reporting timelines (i.e., reporting from the Investigator/site to the Sponsor/designee) in Section 11.11.3.1.

Pregnancy outcomes will be reported on the Pregnancy Outcome Report Form. Pregnancy outcomes include live birth (full term or pre-term), stillbirth, spontaneous abortion, and induced abortion (therapeutic or elective).

If a spontaneous abortion occurs or a congenital abnormality/birth defect is observed (including in utero detections), they must be reported immediately to the Sponsor or designee.

Some pregnancy findings and outcomes (e.g., spontaneous abortion, stillbirth, congenital abnormality/birth defect, [including in utero detections]), may meet seriousness criteria; if so, the SAE should also be reported.

11.12. Reporting of Study Treatment Errors, including Misuse/Abuse

Medication errors are unintentional errors in the prescribing, dispensing, administration or monitoring of a medicine while under the control of a healthcare professional, participant, or consumer.

Misuse refers to situations where the medicinal product is intentionally and inappropriately used not in accordance with the protocol.

Abuse corresponds to the persistent or sporadic, intentional excessive use of a medicinal product, which is accompanied by harmful physical or psychological effects.

Study treatment errors and uses outside of what is foreseen in the protocol will be recorded and reported to the safety database only if associated with an SAE (see Table 7). Misuse or abuse will be collected and reported in the safety database irrespective of it being associated with an AE/SAE within 24 hours of Investigator's awareness.

Table 7: Guidance for Capturing the Study Treatment Errors Including Misuse/Abuse

Treatment error type	Document in AE eCRF	Complete Special Situations Report form
Unintentional study treatment error	Only if associated with an AE	If associated with an SAE
Misuse/Abuse	Only if associated with an AE	Yes, even if not associated with an AE, regardless of seriousness

If an SAE has occurred, refer to Section 11.11.5 for SAE reporting procedures.

11.13. Blood Volume Requirements

Blood sampling for adult participants totals approximately 37 mL across the duration of the study. Total blood sample volume for adolescent participants is approximately 62 mL.

At the Day 1 visit for adolescent participants only, the total blood sampling volume is 28 mL, which is the highest volume at any given visit, equivalent to 1% of the blood volume of a 40 kg adolescent. As this is the first study of deucrictibant for the on-demand treatment of HAE attacks in adolescents, PK sampling is included to support the characterization of the benefit-risk profile in adolescents.

12. STATISTICS

12.1. General Considerations

The analyses of the primary and secondary efficacy endpoints will be primarily based on the Primary Efficacy Analysis Set, with supportive analyses based on the Supportive Efficacy Analysis Set. Safety analysis will be based on Safety Analysis Set. General disposition and baseline characteristic summaries will be based on all enrolled participants.

For descriptive summaries, continuous endpoints will be summarized using descriptive statistics (number of observations [n], mean, standard deviation [SD], median, first quartile [Q1], third quartile [Q3], minimum, and maximum), unless otherwise specified; binary and categorical endpoints will be summarized using frequency and percentage of participants.

Formal statistical testing will be performed on the primary and secondary efficacy endpoints. A gatekeeping procedure will be used to control study-wise type 1 error rate at a 2-sided alpha of 0.05 across all endpoints tested, following the order of first testing the primary endpoint, then the secondary endpoints in the fixed order as pre-defined in Section 5.

The statistical analysis plan (SAP) will be finalized before database lock, and will include a more technical and detailed description of the statistical analyses described in this section.

12.2. Determination of Sample Size

The primary endpoint of the study is time to onset of symptom relief, defined as PGI-C rating of at least “a little better” for 2 consecutive timepoints within 12 hours post-treatment. [redacted], a total of 60 participants in the Primary Efficacy Analysis Set (PEAS) will provide at least 90% power to demonstrate deucrictibant is superior to placebo in reaching time to onset of symptom relief by PGI-C with a 5% significance level for a 2-sided test. Considering that some participants may not experience one study-treated attack in each treatment period, and thus not be included in the PEAS, the study may enroll approximately 120 participants in total to ensure that the requirement of at least 60 participants with 2 study treated attacks (having an attack treated with study drug in each of the 2 attack periods) is fulfilled.

12.3. Analysis Sets

The Primary Efficacy Analysis Set (PEAS) consists of all randomized participants who had an attack treated with study drug in each of the 2 attack periods, namely, those who had 2 attacks treated with study drug.

The Supportive Efficacy Analysis Set (SEAS) is defined as all randomized participants who had at least one attack treated with study drug.

The Safety Analysis Set includes all enrolled participants who received any dose of study drug.

The Pharmacokinetic Analysis Set is defined as all adolescent participants (12 to <18 years at the time of screening) who receive the Day 1 non-attack dose of study drug and have at least 1 measurable concentration of deucrictibant at the Day 1 visit.

12.4. Efficacy Estimands and Analyses

12.4.1. Primary Efficacy Estimand and Analysis

The targeted treatment regimen to be used for this study is deucrictibant (20 mg strength) or placebo for oral use in treating HAE attacks. The primary estimand on efficacy of deucrictibant in treating HAE attacks is for the patient population as defined through the study inclusion and exclusion criteria and currently having an HAE attack, with the primary endpoint as time to onset of symptom relief, defined as PGI-C rating of at least “a little better” for 2 consecutive timepoints within 12 hours post-treatment. The Kaplan-Meier (KM) estimated median and percentiles of time to onset of symptom relief by PGI-C will be provided along with a KM plot for each treatment group.

The intercurrent event (ICE) is the intake of rescue medication or prohibited medication after the attack is treated with study drug. Composite variable strategy will be used to address the ICE in the primary estimand. The occurrence of ICE will be regarded as a treatment failure. Namely, if rescue medication or prohibited medication was taken within 12 hours post-treatment, the time to onset of symptom relief will be censored at the maximum time of consideration (i.e., 13 hours, which is the upper end of the 12-hour time window in the electronic device for participant to record data at that nominal timepoint), regardless whether PGI-C reached the rating of at least "a little better".

For the primary endpoint of time to onset of symptom relief by PGI-C (defined as PGI-C rating of at least “a little better” for 2 consecutive timepoints within 12 hours post-treatment), the [redacted] test will be used to test the null hypotheses that there is no difference in time to onset of symptom relief by PGI-C between the treatment with deucrictibant and with placebo. Supportive analysis will be performed using a marginal Cox proportional hazards model (CPHM) with a robust variance-covariance estimator to account for the within-subject correlation. The model will include treatment (deucrictibant or placebo), attack period (attack 1 or attack 2), and sequence (deucrictibant/placebo or placebo/deucrictibant) as independent variables. The estimated hazard ratio between deucrictibant and placebo will be provided with 95% confidence interval (CI) and p-value.

As supplementary analyses, the above analyses will be performed using a treatment policy strategy, in which the data from all efficacy assessments through 12 hours post-treatment, including those collected after intake of rescue medication or prohibited medication, will be used for analysis on the primary endpoint.

When appropriate, supportive analyses may be performed on the primary endpoint and other selected efficacy endpoints using the Supportive Efficacy Analysis Set based on subgroups by pre-treatment attack characteristics such as attack location, severity, and timing of study drug administration.

12.4.2. Secondary Efficacy Estimands and Analyses

The secondary endpoints in fixed order of testing are presented in Section 5.

For all secondary efficacy endpoints including time-to-event endpoints and binary endpoints, estimands will be defined in a similar way as the primary endpoint, composite variable strategy will be consistently used to handle ICEs for all secondary efficacy endpoints.

For binary secondary endpoints, i.e., secondary efficacy endpoints related to proportions, composite variable strategy will also be used to address ICEs as follows:

- For the secondary endpoint of proportion of study drug-treated attacks achieving a PGI-C rating of at least “a little better” at 4 hours post-treatment, the ICE is again the intake of rescue medication or prohibited medication. Following the composite variable strategy, the ICE is regarded as an unfavorable outcome, that is, the attacks with rescue medication use or prohibited medication use within 4 hours will be regarded as not achieving symptom relief at 4 hours post-treatment (i.e., an outcome of "No" in the binary outcome variable) regardless of whether PGI-C achieved “a little better”. The proportion and 95% CI will be provided for each treatment group together with the p-value from Prescott test (Prescott 1981). Other binary endpoints characterizing attack symptom relief and symptom resolution will the handled similarly.
- For the secondary endpoint of proportion of study drug-treated attacks requiring rescue medication within 24 hours post-treatment, the ICE is the intake of prohibited medication. Following the composite variable strategy, the attacks with intake of prohibited medication within 24 hours post-treatment will be regarded as an unfavorable outcome and counted in the number of attacks requiring rescue medication at 24 hours. The proportion and 95% CI will be provided for each treatment group together with the p-value from Prescott test.
- For binary secondary efficacy endpoints, supplementary analyses will be provided using the Supportive Efficacy Analysis Set and based on the generalized estimating equation (GEE) with a logit link. The model will include treatment (deucrictibant or placebo), attack (attack 1 or attack 2), and sequence (deucrictibant /placebo or placebo/deucrictibant) as independent variables, and an identity matrix as the working correlation matrix of the robust empirical estimates of error (Jones, 2014). The estimated odds ratio between deucrictibant and placebo will be provided with the 95% CI and p-value.

12.5. Analysis of Safety

Safety endpoints will be summarized by actual treatment received using the Safety Analysis Set. Frequencies and percentages will be provided.

The rates of TEAE, Serious TEAE, TEAE leading to discontinuation will be summarized by MedDRA Preferred Terms and by treatment group.

If applicable, descriptive statistics of safety related baseline assessments will be tabulated with those of their repeat assessments during the study (e.g., safety laboratory tests, vital signs, ECG), by visit/time point.

12.6. Pharmacokinetic Analyses

PK concentration and PK parameters will be descriptively summarized based on the PK analysis set.

For adolescent participants, the following PK parameters will be derived from the concentration-time curve:

- Maximum plasma concentration
- Time to peak concentration

The following PK parameters will be calculated:

- Elimination half-life
- Total plasma clearance
- Volume of distribution
- Area under the plasma concentration time curve

12.7. Interim Analysis

Not applicable.

13. DIRECT ACCESS TO SOURCE DATA/DOCUMENTS

13.1. Study Monitoring

The Sponsor is responsible for monitoring the clinical study to ensure that participants' human rights, safety, and well-being are protected, that the study is properly conducted in adherence to the current protocol and any study-specific documents (e.g., eCRF completion guidelines), to Good Clinical Practice (GCP) ICH E6(R2), and to applicable regulatory requirements, and that study data reported by the Investigator are accurate and complete and that they are verifiable with study-related records such as source documents. No data that does not have source documentation will be recorded in the eCRF. The Sponsor or designee will determine the extent and nature of the monitoring and assign sponsor-independent clinical monitors who are GCP-trained. The Sponsor or designee will provide the monitors with adequate training on the study protocol. The study will be monitored in accordance with the clinical study monitoring plan, which specifies the monitoring tasks.

Monitoring visits will be performed to confirm appropriate study conduct. If applicable and permissible per local regulations, monitoring visits may be performed remotely. Monitoring or remote monitoring may include assessment of the following:

- The investigational team is adhering to the study protocol
- Informed consent has been obtained from all participants
- AEs/SAEs are being reported as required
- Data are accurately recorded in the eCRFs
- Study drug is being stored correctly and drug accountability is being performed on an on-going basis
- Facilities are, and remain, acceptable throughout the study
- The Investigator and site are receiving sufficient information and support throughout the study
- eCRFs and Study Records

A validated EDC system will be used for entry of the data by the investigator or other authorized user into study specific eCRFs designed by the sponsor or its designee. All information recorded on eCRFs for this study must be consistent with the participant's source documentation. The Investigator or designee will attempt to obtain specific source documents for all data recorded in the eCRF (e.g., eligibility criteria, medical history). Where there is no specific source available in the medical records, the Investigator or designee will attempt to obtain specific, written source documents from previous treating physicians. Only when no written source is available, may the Investigator or designee obtain responses from the participant him/herself during the intake interview(s), record these into the source documents and enter the data into the eCRF.

Initial data entry and any changes to the data will be made only by authorized users, and data entries and changes will be captured in an electronic audit trail. An explanation of any data change will be recorded.

A validated ePRO system for use on an electronic device will be used for entry of the patient reported outcome data by the participant into a study-specific diary designed by the Sponsor or its designee. Provisions will be made for back-up method(s) in case of device malfunction. The ePRO data and back-up data will be considered source data.

13.2. Audits and Inspections

Authorized representatives of the Sponsor, a regulatory authority, an IEC or an IRB may visit the site to perform audits or inspections, including source data verification. The purpose of a Sponsor audit or inspection is to systematically and independently examine all study-related activities and documents to determine whether these activities were conducted, and data were recorded, analyzed, and accurately reported according to the protocol, GCP guidelines of the ICH, and any applicable regulatory requirements. The Investigator should contact the Sponsor or designee immediately if contacted by a regulatory agency about an inspection.

13.3. Independent Ethics Committee/Institutional Review Board

In accordance with GCP, the protocol, informed consent form(s), recruitment materials, and all participant materials will be submitted to an appropriate IEC/IRB for review and approval. Approval of both the protocol and the ICF must be obtained before any participant is enrolled. Any substantial amendment to the protocol will require review and approval by the IEC/IRB before the changes are implemented to the study. Non-substantial amendments, e.g., containing administrative changes, will only be submitted to the IEC/IRB for notification. All changes to the consent form must be IEC/IRB approved, in accordance with local regulations.

13.4. Data Monitoring Committee

An Independent Data Monitoring Committee (IDMC) will be assembled before the start of the study to review safety data on a regular basis (and on an ad hoc basis in case of unexpected safety issues) throughout the study. The IDMC charter will be prepared before the start of the study and will identify the IDMC membership and outline the expected review of unblinded safety data. The IDMC will consist of independent experts external to the study.

The responsibilities of the IDMC include, but are not limited to:

- To minimize the exposure of participants to an unsafe therapy or dose
- To make recommendations for changes in study processes where appropriate
- To recommend continuation of the study if warranted

Further details on the membership, responsibilities and working procedures of the IDMC are described in the IDMC charter, provided as a separate document in the study file.

14. QUALITY CONTROL AND QUALITY ASSURANCE

Procedures relevant to the clinical management of this study are either described in the applicable standard operating procedures (SOPs) of Pharvaris or their representatives. In accordance with Pharvaris SOPs, this study may be subject to independent quality assurance (QA) GCP audits at the study sites.

Pharvaris will not allow any waivers to the protocol throughout the conduct of the study. A deviation from the protocol (not implemented in an amendment or intended as a systematic change) that could not be justified as necessary to protect the safety, rights, or welfare of the participants, will be reported and followed up according to the local IEC/IRB requirements.

Investigators and other study personnel will be appropriately trained by the Sponsor or its designee. The study will be monitored (see also Section 13.1).

To ensure the collection of accurate, consistent, and reliable data, safety laboratory tests will be done at a central laboratory. Based on Investigator determination or local requirement for participant safety, additional samples may be assessed during the study by a local laboratory, in parallel with the central laboratory.

Serious breaches will be reported in accordance with EU/European Economic Area Regulation (EU) No 536/2014, Article 52:

- The Sponsor or designee shall notify the Member States concerned about a serious breach or of the version of the protocol applicable at the time of the breach through the EU portal without undue delay but not later than 7 days of becoming aware of that breach.

A serious breach is defined as a breach likely to affect to a significant degree the safety and rights of a participant or the reliability and robustness of the data generated in the clinical study.

15. ETHICS

15.1. Ethical Conduct of the Study

The study will be conducted in accordance with this study protocol, the International Council for Harmonisation GCP (ICH GCP E6[R2]), 536/2014 EU CTR, and other applicable local laws and regulations. Compliance with ICH-GCP provides public assurance that the rights, safety, and well-being of study participants are protected, consistent with the principles that have their origin in the Declaration of Helsinki, and that the clinical study data are credible.

The Investigator will assure that no deviation from, or changes to the protocol will take place without prior agreement from the sponsor and documented approval from the IEC/IRB, except where necessary to eliminate an immediate hazard to the participants. All relevant personnel involved in the conduct of this study will have completed ICH GCP training.

Participant insurance will be provided according to local regulations.

15.2. Written Informed Consent

- At the screening visit, a signed ICF/assent must be obtained from each participant and/or parent/legally designated representative/guardian before performing any study-related procedures. Each participant will be given both verbal and written information describing the nature and duration of the clinical study.
- The ICF must describe in detail the study drug, study procedures, and risks/potential burden of study participation. The ICF and any other written study material provided to the participant must be approved by an IEC/IRB before participant enrollment.
- The Investigator or designee is responsible for explaining the nature and purpose of the study as well as other study-related matters to participants, using the written information, and for obtaining their full understanding and written consent to participate in the study at their own free will. No participant may be subjected to undue influence, such as compulsory enrollment into a study. The Investigator or designee who provided explanations to the participant should sign and date the ICF as well as the participant. The participant should receive a copy of the signed ICF (or a duplicate original if required by local regulations) and an original must be filed by the Investigator. The process and communication of the consent should be documented on medical records of the participant. The signed ICFs will be retained by the investigator and made available upon request (for review only) to the study monitor and auditor.
- The language and expressions used in the written information should be as plain and understandable as possible. Participants should be given the opportunity to ask questions and receive satisfactory answers to the inquiry and should have adequate time to decide whether or not to participate in the study. Written information should not contain any language or content that causes the participant to waive or appears to waive any legal rights, or that releases/mitigates or appears to release/mitigate the study site, the investigator, other study personnel, collaborators, or the Sponsor from liability for negligence.

- For adolescent participants (≥12 to <18 years of age) the language and expressions used in the written assent and other information should be age appropriate and a separate written consent should be provided to the participant's parent/legally designated representative/guardian, in accordance with local regulations.
- If an adolescent participant reaches the age of 18 years or as determined by local law during study participation, the participant will be required to provide written informed consent in order to continue study participation, in accordance with local regulations.
- If an ICF is updated as a result of a substantial protocol amendment, the new IEC/IRB approved version will be used to re-consent currently enrolled participants and must be provided to newly enrolled participants before their entry into the study.

15.3. Participant Confidentiality

The conduct of this study and the processing of any personal data collected from each subject (or from a subject's healthcare professional or other relevant third-party sources) by the Sponsor or it's designee, the site and the Investigator for use in the study will fully adhere to the requirements set out in applicable data protection and medical privacy laws or regulations, including, without limitation, the General Data Protection Regulation (GDPR) EU 2016/679. The Sponsor or its designee shall ensure that at all times it has an appropriate legal basis for processing personal data under applicable data protection law. Site-based organizational and technical arrangements to avoid unauthorized access vary by site but all include access-controlled/access-limited document control and technical solutions including passwords and security control measures to protect study-specific data, both in paper and electronic format.

The investigators shall provide coded data to the Sponsor or its designee, which does not reveal the participant's name, full date of birth, or any other information which can identify the participant. All personal information shall be replaced with a Subject Identification Code before any information leaves the investigator sites.

The Investigator shall report any data breaches that might occur to the Sponsor or its designee, without undue delay. The Sponsor has implemented a Business Practice to address Data Breaches that complies with the requirements of applicable laws and regulations including the GDPR. The Data Breach procedures in the Business Practice provide specific responses to actual or potential threats and involve investigation, containment and mitigation. If applicable, the authorities and the data subjects shall be notified of a data breach, within the required timeframes of the applicable laws and regulations, including those of the GDPR.

The Sponsor will use the information obtained during the conduct of this study for the development of deucrictibant soft capsule. The Investigator is obliged to provide the Sponsor or its designees with complete test results and all data collected in this study, as described in this protocol.

Access to stored samples will be limited. Samples and data will be stored using codes assigned by the sponsor. The code key linking the participant name to the research records will be kept by the Investigator and may not be released outside the study site. The coded data does not include any identifying information for the participant, such as name or address. Data will be kept in

password-protected computers. Only Investigators, the Sponsor and affiliates, or contracted parties will have access to the coded samples and data.

Data generated by this study must be available for inspection by representatives of the European Medicines Agency, the FDA, Health Canada, any other competent authority, and the IRB/IEC for the study sites. Even though individuals involved in the study, such as study monitors and auditors, may get to know matters related to participants' privacy due to direct access to source documents, or from other sources, they may not disclose these contents to third parties.

16. DATA HANDLING AND RECORDKEEPING

16.1. Inspection of Records

The Investigator and the study site must accept monitoring and auditing by the Sponsor or their representatives as well as inspections from the IEC/IRB and relevant regulatory authorities. Source documents may be reviewed remotely using regulatory-compliant and documented computerized systems when agreed upon with the investigator and study site, as applicable by local regulations. In these instances, they must provide all study-related records, such as source documents when they are requested by the monitors and auditors, the IEC/IRB, or regulatory authorities. The confidentiality of the participants' identities shall be well protected consistent with local and national regulations when the source documents are subject to direct access. In accordance with the sponsor's procedures, this study may be subject to an independent QA GCP audit at the study sites.

16.2. Retention of Records

All records (data and documents) relating to the study that are generated by or collected by the contract research organization (CRO) will be filed in an eTMF according to ICH-GCP. The eTMF will be available for sponsor review at any time and will be transferred to the sponsor after the study is complete and will be archived for at least 25 years or as otherwise required by local regulations. Also, all records relating to the study that are generated by or collected by the investigators will be filed in an investigator site file according to ICH GCP. Investigators will retain the investigator site file for at least 25 years or as otherwise required by local regulations. The Investigators will also retain source documents under their control for at least 25 years or as otherwise required by local regulations. All medical records will be retained by the medical center according to their own internal procedures.

16.3. Protocol Deviations

Protocol deviations will be recorded based on the Protocol Deviation Plan. The following deviations from the protocol will be considered protocol violations:

- Deviations from the inclusion and exclusion criteria
- Deviations from concomitant medication restrictions
- Deviations from any other protocol requirements that result in a significant added risk to the participant or have an impact on the quality of the data collected or the outcome of the study.

Protocol deviations will be reported in the final Clinical Study Report.

16.4. Data Generation and Analysis

The Investigator or designee must enter all protocol-required data in the available eCRF, which is a validated system. In the interest of collecting data in the most efficient manner, the Investigator or designee should enter data into the eCRF as soon as possible after the participant's visit. Laboratory results from the central laboratory will be electronically transferred. The eCRFs and any supporting documents should be available for retrieval at any

given time. The Monitor should verify the data in the eCRFs with source documents and confirm that there are no inconsistencies between them. If any inconsistency is detected on the eCRFs, the Monitor must query the Investigator and the Investigator or designee should make corrections/additions in the eCRF or provide an explanation within the eCRF system. The Monitor should verify the corrected data with source documents to confirm that there are no inconsistencies between them, and also check that appropriate records on the corrections/additions of data are maintained.

Electronic patient reported outcomes (ePRO) data will be collected via provisioned electronic device. Participants should carry the electronic device with them at all times in order to collect information on symptoms and details of HAE attacks. The electronic device is a validated 21 CFR Part 11-compliant electronic data capture tool. Provisions will be made for back-up method(s) in case of device malfunction. Data from these devices will be transmitted immediately to the ePRO vendor database when the device is online, otherwise, it will be transmitted as soon as the device is connected to the internet. The data will be reviewed by investigators and site staff to ensure compliance with timepoint expectations and to ensure that data is clinically reasonable. If there are questions, the site staff will reach out to the participant and will request clarification of the data entered. If there is a change required, the site staff will document the original answer in the participant record and also document the participant confirmed change that is required. Site will then use this information to trigger a Data Clarification Request within the ePRO vendor system so that corrections that have been confirmed with the participant may be made to the reported data.

For screening failures, defined as participants who signed informed consent but were not eligible for study participation, relevant data will be collected in the eCRF.

The Investigator is responsible for ensuring that all data in the eCRFs are accurate and complete, and that all entries are verifiable with source documents. Source data must be available at the site to document the existence of the study participants, and to substantiate the integrity of study data collected. In case a participant is prematurely discontinued, the reason for discontinuation must be documented in the source documents and eCRF. Study data collected from a participant before study discontinuation will still be used in the overall data analysis.

Data management will be coordinated by a delegated CRO, in accordance with their SOPs for data management. All study specific processes and definitions will be described in the data management plan. Coding of medical terms will be performed using MedDRA and medications will be coded with the World Health Organization-Drug Dictionary.

16.5. Financial Disclosure

Any Investigator involved in conducting the study is required to disclose any financial arrangement during the study and for 1 year after, whereby the outcome of the study could be influenced by the value of the compensation for conducting the study, or other payments the investigator received from the Sponsor. The following information is collected: any significant payments from the Sponsor or subsidiaries such as a grant to fund ongoing research, compensation in the form of equipment, retainer for ongoing consultation or honoraria; any proprietary interest in the study drug; any significant equity interest in the Sponsor or

subsidiaries as defined in 21 CFR 54.2(b). A financial disclosure form is to be signed by each Investigator.

17. PUBLICATION POLICY

All data generated from this study are the property of the Sponsor and shall be held in strict confidence along with all information provided by the sponsor. Except as provided through written agreement between the Sponsor and the Investigator, independent analysis and/or publication of these data by the Investigator or any member of his/her staff is not permitted without prior written consent of the sponsor. Such consent will not be withheld unreasonably. The Sponsor is in agreement with the principle of full disclosure of clinical study results.

Within one year from the end of a clinical study in all Member States concerned, the sponsor will submit to the EU database CTIS a summary of the results of the clinical study. The content of the summary of the results is set out in CTR Annex IV. It shall be accompanied by a summary written in a manner that is understandable to laypersons.

18. LIST OF REFERENCES

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19. APPENDICES

APPENDIX 1. HAEA AND WAO/EAACI GUIDELINES

HAE-nL-C1INH Sub-type	Gene/Underlying mutation	Supporting Literature
HAE-FXII	F12, the gene encoding coagulation FXII	Cichon, 2006
HAE-PLG	PLG, the gene encoding plasminogen	Bork, 2018
HAE-ANGPT1	ANGPT1, the gene encoding angiopoietin-1	Bafunno, 2018
KNG1	kininogen 1 gene	Bork, 2019
HAE-MYOF	Myoferlin gene	Ariano, 2020
HAE-HS3ST6	Heparan sulfate 3-O-sulfotransferase 6 gene	Bork, 2021
HAE-unknown (HAE-U or HAE-UNK)	For whom the responsible mutation has not yet been defined	N/A

Sources: Busse, 2021 and Maurer, 2022

APPENDIX 2. SUMMARY OF CHANGES

Amendment 1.2 (dated 03 July 2024) was amended to Amendment 4.1 (dated 17 January 2025).

The primary reasons for the amendment were to allow enrollment of participants with HAE type 3. This change is intended to help address an unmet medical need for the development of an effective oral treatment for this type of HAE. Also, based on European Medicines Agency Paediatric Committee feedback, it was stipulated that study drug treatment of qualifying non-severe laryngeal attacks will be permitted in adult participants only.

Substantive changes to the protocol included the following:

- The text was updated to indicate that the study will include participants with HAE type 3 (Section 6.1 Overall Study Design, Section 7 Selection and Withdrawal of Participants).
- The study inclusion/exclusion criteria were revised to allow enrollment of participants with HAE type 3.
  - Inclusion criterion 3b (Section 7.1 Participant Inclusion Criteria)
  - Exclusion criterion 2 (Section 7.2 Participant Inclusion Criteria)
- HAEA and WAO/EEACI Guidelines were cited to support diagnosis of HAE type 3 (Section 4.1 Hereditary Angioedema, Section 7.1 Participant Inclusion Criteria, Appendix 1 HAEA and WAO/EAACI Guidelines).
- The list of references was updated to include sources included in the HAEA and WAO/EAACI Guidelines (Section 18 List of References).
- The protocol text and list of references were updated to cite scientific articles describing laboratory tests that can be used to support identification of HAE type 3 patients (Section 4.1 Hereditary Angioedema, Section 18 Section 4.1 List of References).
- The text was revised to indicate that non-severe laryngeal attacks qualify for study drug treatment in adult participants only (Section 6.1 Overall Study Design, Section 6.1.2.3.1 Attack Qualification and Treatment, based on Attack Type, Section 6.1.2.3.1.2 Laryngeal Attacks in Adolescents or Laryngeal Attacks in Adults with Breathing Difficulties or Stridor, Section 6.1.2.3.1.3 Non-Severe Laryngeal Attacks without Breathing Difficulties or Stridor (in Adults Only).
- In alignment with the inclusion criteria for HAE type 3 participants, it was clarified in the study schedule of events and study activities that for participants with HAE type 3, diagnostic testing results are to confirm C1-INH function ≥50% of normal and C4 level not below lower level of normal range performed by the central laboratory (Section 6.1.1 Schedule of Events Table 4 and Table 4 Footnote h, Section 6.1.2.1 Screening Phase (Days -28 to -1).
- Due to the requirement for C4 diagnostic testing of HAE type 3 participants, the total blood sampling across the duration of the study for adults and adolescents was

revised to approximately 37 mL and 62 mL, respectively (Section 11.13 Blood Volume Requirements).

- A typographical error in the study schedule of events table header for the End-of-Study Visit was corrected to read "10 days (±5) after attack resolution" (Section 6.1.1 Schedule of Events Table 4) in alignment with Table 4 Footnote c.

In addition, changes made in Amendment 4.1 included several minor editorial and administrative changes. The protocol synopsis was updated in alignment with changes made to the protocol body.

NOTE: the current amendment also incorporates changes made in prior global protocol amendments: Amendment 3.0 (dated 28 October 2024) and Amendment 2.0 (dated 19 May 2024).

Substantive changes in Amendment 3.0 included the following:

- The timing of the End-of Study Visit after the second qualifying attack was clarified (Section 6.1, Section 6.1.2.4.2).
- The procedure for C1-INH and C1q testing was clarified (Section 6.1.2.1, Section 7.1).
- The procedures for ePRO retraining and repeat assessments were clarified (Section 6.1.2.2, Section 6.1.2.3.1.1).
- The procedure for attack qualification was clarified (Section 6.1.2.3.1.1, Section 6.1.2.3.1.3, Section 10).
- It was noted that adolescent participants (≥12 to <18 years of age) are to be enrolled in accordance with local age requirements (Section 7).
- Exclusion criterion 4 regarding use of attenuated androgens was modified (Section 7.2).
- Language regarding concomitant use of CYP3A4 inhibitors/inducers was clarified (Section 7.2, Section 8.2.4.1).
- It was clarified that participants will not be replaced in this study (Section 7.3.1).
- Contraceptive language was updated per regulatory feedback and applicable regulations (Section 8.2.3).
- Several sections were updated to reflect current standard Sponsor language and based on regulatory agency feedback:
  - Benefit-risk language was augmented (Section 4.4.1).
  - It was clarified that the Sponsor undertook patient involvement in the design of the study in multiple global patient forums, which informed design aspects with feedback to reduce participant burden where possible (Section 6.2).
  - The SAE definition was clarified (Section 11.11.1.2).
  - The definition of clinically significant abnormalities was clarified (Section 11.11.1.3).

  - Previous Section 11.11.1.3 (Expectedness) and 11.11.1.4 (Clinical Laboratory Adverse Events) were removed.
  - The criteria for assessment of AE relationship to study drug were clarified (Section 11.11.2).
  - The procedures for reporting AEs were clarified (Section 11.11.3).
  - The timing of AE collection was clarified (Section 11.11.3.1).
  - The criteria for assessment of AE severity were clarified (Section 11.11.3.2).
  - The procedure for AE follow-up was added (Section 11.11.4).
  - Procedures for reporting adverse events were clarified (Section 11.11.5).
  - Procedures for reporting SUSARs were added (Section 11.11.6).
  - Procedures for reporting study treatment errors, including misuse/abuse, were clarified (11.12).
  - Section 11.13, Blood Volume Requirements, was added.
- The procedure regarding ECG assessment and recording was clarified (Section 11.6).
- The procedure for clinical laboratory assessment, including safety and pregnancy testing, was clarified to allow for more frequent pregnancy testing as per local requirements and also allow for laboratory tests to be assessed locally in parallel with the central laboratory (Section 11.7, Section 14).
- Details regarding sample storage and retention were removed, as they are addressed in the Laboratory Manual (Section 11.7.1 Section 11.8).
- Handling of on-study pregnancies was clarified (Section 11.11.7).
- Details regarding the handling of serious data breaches were added (Section 14).
- Details regarding attainment of written informed consent from adolescent participants were clarified (Section 7.1, Section 11.1, Section 15.2).
- It was clarified that the Safety Analysis Set will include all enrolled (rather than randomized) participants who received any dose of study drug (Section 12.3). Accordingly, general disposition and baseline characteristic summaries also will be based on all enrolled (rather than randomized) participants (Section 12.1).

Substantive changes in Amendment 2.0 included the following:

- The study population was revised to include approximately 30% of participants who are using long-term prophylactic medication for HAE globally; in the United States (including Puerto Rico), enrollment is targeted to be approximately 50% of participants using long-term prophylactic medication (Section 8.2.2, Section 8.4). The inclusion criteria were revised accordingly (Inclusion criterion 6, Synopsis, Section 7.1).
- After the on-site Screening Visit, it was clarified that the Investigator will confirm whether the participant is eligible for the study after the results of all safety laboratory tests have become available (Section 6.1.2.1).

- If a participant opts to have their Day 1 visit performed remotely, it was clarified randomization should take place one day before the Day 1 visit, so that study drug can be shipped to the participant's home in time for the Day 1 visit (Section 6.1.2.2).
- Inclusion criterion 3 was modified to indicate that if treatment washout is required, the confirmatory C1-INH testing should be performed at least 5 half- lives after the last dose of C1-INH or attenuated androgens therapy (Synopsis, Section 7.1).
- Inclusion criterion 9 was clarified to indicate that female participants of child-bearing potential are to use acceptable methods of birth control as defined in the protocol and as available locally (Synopsis, Section 7.1, Section 8.2.3).
- It was clarified that use of concomitant medications ***with systemic absorption*** that are strong inhibitors/inducers of CYP3A4, such as clarithromycin, itraconazole, ketoconazole, and ritonavir within the last 30 days, or within 5 half-lives (whichever is longer) of the time of randomization are a criterion for exclusion (Criterion 11) and are prohibited during the study.
- The window for the AMRA interview was increased from 48 to 72 hours after the Day 1 visit (Section 6.1.1, Section 10.5).